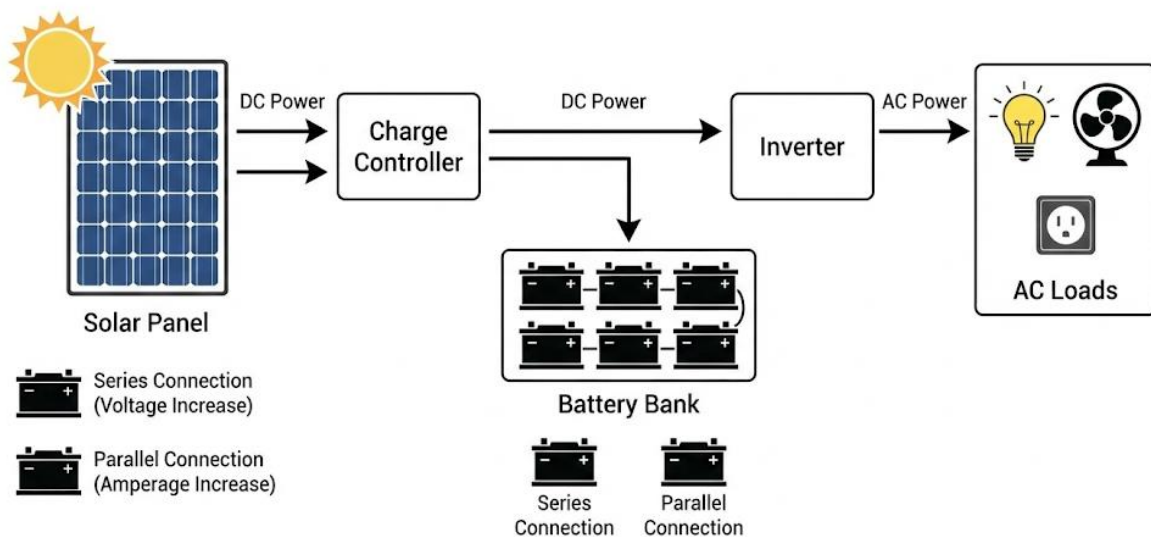


SS2 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF SS1 WORK

BASIC OFF-GRID SOLAR SYSTEM



1.1 INTRODUCTION

Welcome to SS2 Solar Photovoltaic Installation. This term, we will be building upon the foundational knowledge from SS1, moving from basic concepts into the design and regulation of more advanced and larger-scale solar systems. This first week is dedicated to a thorough review to ensure everyone has a strong grasp of the fundamental principles.

1.2 RECAP OF SOLAR BASICS

1. What is Solar Energy?

- Solar energy is radiant light and heat from the sun that is harnessed using a range of technologies. It is a renewable, inexhaustible, and clean source of energy.

2. The Photovoltaic (PV) Effect:

- This is the core principle of all solar panels. It is the process by which a material (typically silicon) generates a direct current (DC) of electricity when exposed to sunlight (photons).
- When photons strike a solar cell, they knock electrons loose from their atoms. These free-flowing electrons are channeled into a current, which is electricity.

3. **Direct Current (DC) vs. Alternating Current (AC):**

- **DC:** The type of electricity produced by solar panels and stored in batteries. Electrons flow in one constant direction.
- **AC:** The type of electricity used in our homes and supplied by the grid (NEPA/PHCN). Electrons rapidly alternate their direction.

1.3 REVIEW OF CORE COMPONENTS

A basic standalone (off-grid) solar system consists of four main components:

1. **SOLAR PANELS (PV MODULES):**

- **Function:** To convert sunlight into DC electricity via the photovoltaic effect.
- **Types:**
 - **Monocrystalline:** Made from a single silicon crystal. Highest efficiency (15-22%), most expensive, black-ish tint, long lifespan.
 - **Polycrystalline:** Made from multiple silicon fragments melted together. Lower efficiency (13-16%), less expensive, blue-ish tint, good lifespan.
 - **Thin-Film:** Made from non-crystalline silicon or other materials. Very low efficiency, cheapest, flexible, but requires a lot of space.

2. **CHARGE CONTROLLER:**

- **Function:** The "brain" of an off-grid system. It regulates the voltage and current coming from the solar panels to the batteries.
- **Importance:** It prevents battery **overcharging** (which boils the battery) and **deep discharge** (which kills the battery).
- **Types:**
 - **PWM (Pulse Width Modulation):** An older, "switch-based" technology. Less efficient, cheaper. Must match panel voltage to battery voltage.

- **MPPT (Maximum Power Point Tracking):** A modern, "smart" technology. Uses a converter to find the "sweet spot" of the panel's voltage and current. 20-30% more efficient than PWM. Allows for higher voltage panel arrays.

3. BATTERY BANK:

- **Function:** To store the DC electricity produced by the panels for use at night or on cloudy days.
- **Type (SS1 Focus):** Deep-Cycle Lead-Acid batteries (Flooded, AGM, Gel). These are designed to be discharged and recharged many times, unlike a car battery.

4. INVERTER:

- **Function:** To convert the low-voltage DC electricity from the batteries into high-voltage AC electricity that can power household appliances.
- **Types:**
 - **Off-Grid Inverter:** Requires batteries to function.
 - **Grid-Tied Inverter:** Does not use batteries; synchronizes with the grid.
 - **Hybrid Inverter:** Can manage panels, batteries, *and* the grid.

1.4 REVIEW OF BASIC WIRING

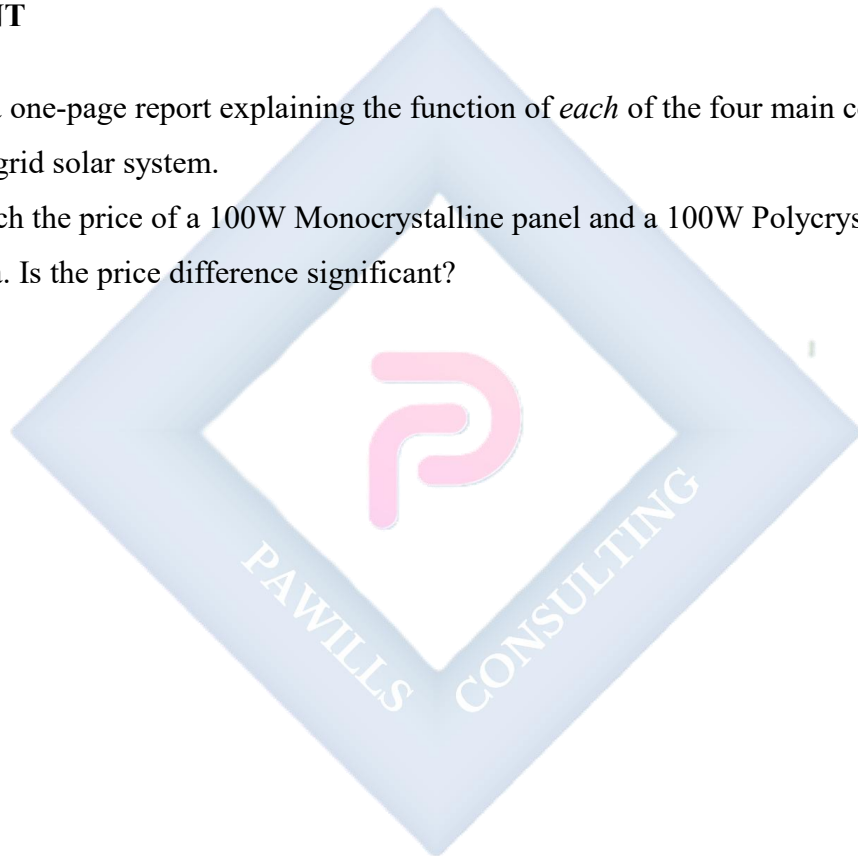
- **Series Connection:** Connecting components positive-to-negative (+ to -).
 - **Voltage Adds Up.**
 - **Amperage (Current) Stays the Same.**
 - *Use:* To increase the voltage of a solar array to meet inverter or charge controller requirements (e.g., two 12V panels in series = 24V).
- **Parallel Connection:** Connecting components positive-to-positive (+ to +) and negative-to-negative (- to -).
 - **Voltage Stays the Same.**
 - **Amperage (Current) Adds Up.**
 - *Use:* To increase the total current (power) of an array while keeping the voltage the same.

EVALUATION

1. Describe the Photovoltaic Effect in your own words.
2. What is the primary function of a charge controller?
3. Explain the difference between a PWM and an MPPT charge controller.
4. You have four (4) 12V, 5A panels. How would you connect them to get a 24V, 10A output? (Draw a diagram).
5. What are the three main types of solar panels, and what is a key advantage of Monocrystalline?

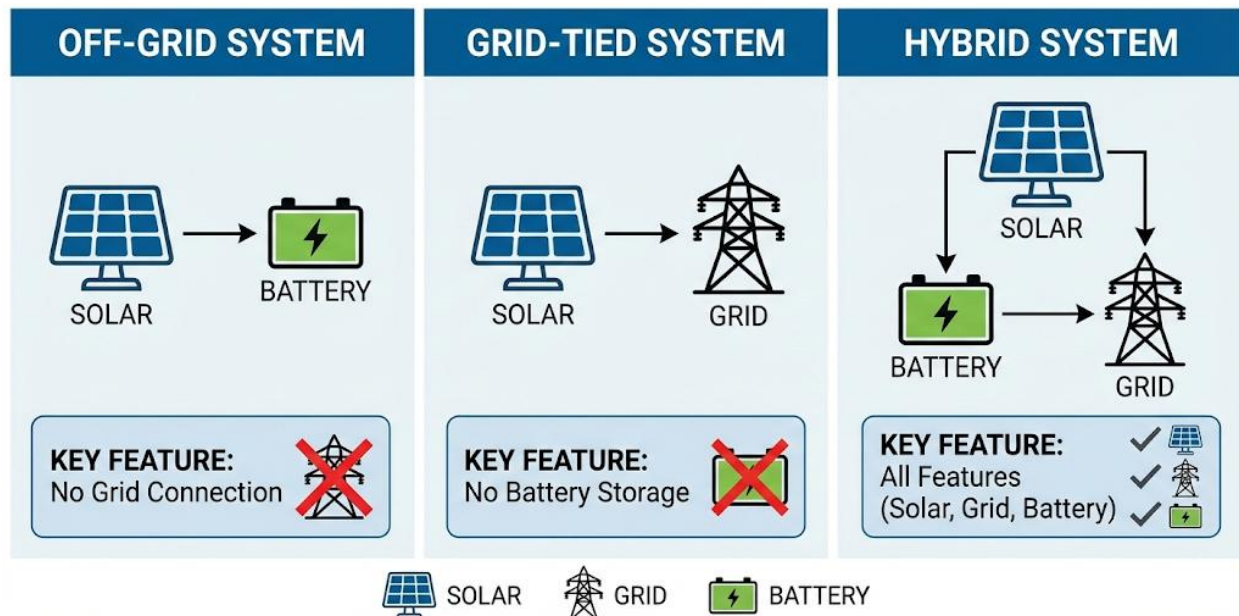
ASSIGNMENT

1. Write a one-page report explaining the function of *each* of the four main components of an off-grid solar system.
2. Research the price of a 100W Monocrystalline panel and a 100W Polycrystalline panel in Nigeria. Is the price difference significant?



WEEK 2: ADVANCED PV SYSTEM DESIGN

ADVANCED PV SYSTEM DESIGN: COMPARATIVE FLOWCHART



2.1 INTRODUCTION

In SS1, we focused on simple off-grid (standalone) systems. In the real world, systems are often more complex, interacting with other power sources. This week, we explore **Hybrid** and **Grid-Tied** systems.

2.2 SYSTEM TOPOLOGIES (TYPES)

1. **Off-Grid (Standalone):** No connection to the utility grid (NEPA). Relies 100% on solar panels and a battery bank. Used in remote areas.
2. **Grid-Tied (On-Grid):** Connected to the utility grid. **No batteries.**
 - **How it works:** During the day, the house is powered by solar. If the panels produce *more* power than the house needs, the excess power is *exported* to the grid (spinning the meter backward - this is called **Net Metering**). At night, the house draws power from the grid as normal.
 - **Pros:** Cheapest system (no batteries), efficient.
 - **Cons:** **Does not work when the grid is down.** (This is a safety feature called "anti-islanding").

3. **Hybrid Systems:** The most advanced and flexible. It combines multiple power sources.

- **Solar + Grid (Grid-Tied with Battery):** This is the most common hybrid. It operates like a grid-tied system but also has a battery bank for *backup*.
 - **Daytime (Grid On):** Powers the house, charges the battery, excess can be exported to the grid.
 - **Night (Grid On):** Powers the house from the battery first, then from the grid.
 - **Day/Night (Grid Off):** The system disconnects from the grid and powers the house from the batteries/panels (like an off-grid system).
- **Solar + Generator:** A common off-grid hybrid. The solar/battery system is the primary source. A diesel/petrol generator is used as a backup to charge the batteries during long periods of poor weather.

2.3 ADVANCED SYSTEM SIZING AND CALCULATIONS

This is the most critical part of an installer's job. Sizing a system involves a step-by-step calculation of the customer's needs.

SCENARIO: A client wants a hybrid system for a small office.

STEP 1: CALCULATE THE LOAD

First, find the total daily energy consumption in **Watt-hours (Wh)**.

Appliance	Quantity	Power (Watts)	Hours/Day	Total Wh/Day
LED Lights	10	10W	8	$10 \times 10W \times 8h = 800 \text{ Wh}$
Laptops	4	60W	8	$4 \times 60W \times 8h = 1,920 \text{ Wh}$
Ceiling Fans	2	80W	8	$2 \times 80W \times 8h = 1,280 \text{ Wh}$
Water Dispenser	1	100W	4	$1 \times 100W \times 4h = 400 \text{ Wh}$
Total				4,400 Wh/day

- **Total Daily Energy Need = 4,400 Wh**

- **Peak Load (all on at once)** = $(10 \times 10) + (4 \times 60) + (2 \times 80) + 100 = 100 + 240 + 160 + 100 = 600 \text{ W}$

STEP 2: SIZE THE INVERTER

The inverter must handle the *maximum power (Watts)* the client might draw at one time (Peak Load).

- **Rule:** Inverter Size = Peak Load x 1.25 (for safety/surge)
- **Calculation:** $600 \text{ W} \times 1.25 = 750 \text{ W}$
- **Conclusion:** We must select an inverter rated for *at least* 750 W. A 1000W (1kW) inverter is a standard, safe choice.

STEP 3: SIZE THE BATTERY BANK

The battery bank must store enough energy (Wh) for use at night or during outages.

- **System Voltage:** Choose a system voltage (12V, 24V, or 48V). Higher voltage is more efficient. For a 1kW system, 24V is suitable.
- **Days of Autonomy:** How many days should the system run with no sun? (Standard = 1 day).
- **Depth of Discharge (DoD):** How much of the battery can be used?
 - Lead-Acid: 50% DoD (to protect battery life)
 - Lithium-ion: 80-90% DoD
- **Calculation (Using Lithium, 80% DoD):**
 - Total Wh needed = (Daily Energy x Days of Autonomy) / DoD
 - Total Wh needed = $(4,400 \text{ Wh} \times 1) / 0.80 = 5,500 \text{ Wh}$
- **Convert Wh to Amp-hours (Ah) at our system voltage:**
 - $\text{Ah} = \text{Wh} / \text{System Voltage}$
 - $\text{Ah} = 5,500 \text{ Wh} / 24\text{V} = 229 \text{ Ah}$
- **Conclusion:** We need a 24V, 229Ah battery bank. (e.g., two 12V, 230Ah batteries in *series*).

STEP 4: SIZE THE SOLAR ARRAY (PV ARRAY)

The array must be large enough to recharge the battery and power the loads *every day*.

- **Peak Sun Hours (PSH):** The average number of hours per day that the sun is at its peak intensity. In Nigeria, this is excellent, averaging **4.5 to 6.5 PSH**. We'll use a conservative 5 PSH.
- **Efficiency Losses:** We lose power from heat, dust, wiring, etc. (Assume 80% efficiency).
- **Calculation:**
 - Total Watt-peak (Wp) of Panels = Daily Energy / (PSH x Efficiency)
 - Total Wp = 4,400 Wh / (5 PSH x 0.80)
 - Total Wp = 4,400 / 4 = 1,100 Wp
- **Conclusion:** We need a **1,100 Wp** (1.1kW) solar array.
- **Panel Selection:** We can buy four 300W panels (1,200 Wp total).

FINAL SYSTEM DESIGN:

- **Inverter:** 1kW, 24V Hybrid
- **Battery Bank:** 24V, 230Ah (e.g., 2 x 12V 230Ah Li-ion)
- **PV Array:** 1.2kWp (e.g., 4 x 300W panels)

EVALUATION

1. What is the main difference between a grid-tied system and a hybrid system?
2. What is "Net Metering"?
3. Why does a grid-tied inverter shut down when the grid fails? What is this feature called?
4. Why is "Peak Sun Hours" (PSH) a critical value in solar calculations?
5. Why do we size an inverter larger than the "Peak Load"?

ASSIGNMENT

1. Using the methodology from this lesson, calculate the Inverter Size, Battery Bank (Ah), and PV Array (Wp) for the following "Load Profile":
 - Total Daily Energy Need: 6,000 Wh

- Peak Load: 1,500 W
- System Voltage: 48V
- Battery Type: Lead-Acid (50% DoD)
- Days of Autonomy: 1
- Peak Sun Hours: 5



WEEK 3: SOLAR ENERGY STORAGE TECHNOLOGIES

3.1 INTRODUCTION

The battery is often the most expensive and most frequently replaced component of an off-grid or hybrid system. The technology has evolved far beyond the standard lead-acid batteries.

Understanding advanced storage is crucial.

SOLAR ENERGY STORAGE TECHNOLOGIES: LEAD-ACID VS. LITHIUM-ION (LFP)			
 LEAD-ACID BATTERIES		 LITHIUM-ION (LFP) BATTERIES	
	CYCLE LIFE LOWER (approx. 300-500 cycles at 50% DoD)		CYCLE LIFE HIGHER (approx. 2000-5000+ cycles at 80% DoD)
	DEPTH OF DISCHARGE (DoD) LIMITED (typically 50% usable capacity)		DEPTH OF DISCHARGE (DoD) HIGHER (typically 80-90% usable capacity)
	COST LOWER UPFRONT, HIGHER LIFETIME (more replacements)		COST HIGHER UPFRONT, LOWER LIFETIME (fewer replacements)
	WEIGHT HEAVIER (bulky and dense)		WEIGHT LIGHTER (compact and less dense) ✨

3.2 RECAP: DEEP-CYCLE LEAD-ACID

- **How it works:** Lead plates submerged in a sulphuric acid electrolyte.
- **Types:**
 - **Flooded (FLA):** Plates are exposed to liquid electrolyte. Must be refilled with distilled water. High maintenance.
 - **Sealed (AGM & Gel):** Maintenance-free.
 - **AGM (Absorbent Glass Mat):** Electrolyte is held in glass mats.
 - **Gel:** Electrolyte is mixed into a gel.
- **Pros:** Low cost, widely available.

- **Cons:** Very heavy, low **Depth of Discharge (DoD)** (max 50%), short **Cycle Life** (500-1000 cycles), sensitive to high temperatures, requires good ventilation (gassing).

3.3 ADVANCED BATTERIES: LITHIUM-ION (LI-ION)

This is the new standard for modern solar installations. The most common type for solar is **Lithium Iron Phosphate (LiFePO₄ or LFP)**.

- **How it works:** Uses lithium ions moving between a positive (cathode) and negative (anode) electrode.
- **Why LFP is better than other Li-ion (like phone batteries):**
 - LFP (Lithium Iron Phosphate) is **thermally stable** and **safer**. It does not catch fire easily, unlike NMC (Nickel Manganese Cobalt) batteries used in phones and EVs.
- **Pros:**
 - **High DoD:** Can be discharged 80-100% without damage.
 - **Long Cycle Life:** 3,000 - 7,000 cycles (5-10 times longer than lead-acid).
 - **High Energy Density:** Much lighter and smaller for the same capacity.
 - **High Efficiency:** 95-98% round-trip efficiency (vs. 80-85% for lead-acid).
 - **Maintenance-Free.**
- **Cons:**
 - **High Upfront Cost:** Significantly more expensive than lead-acid.
 - **Requires a BMS:** Cannot be used without one.

3.4 THE BATTERY MANAGEMENT SYSTEM (BMS)

A BMS is an "electronic brain" attached to a lithium-ion battery. It is **not optional**.

- **Functions of a BMS:**
 1. **Protection:** Prevents the battery from operating outside its safe limits.
 - **Over-voltage (over-charging):** Disconnects the battery.
 - **Under-voltage (over-discharging):** Disconnects the battery.
 - **Over-temperature:** Shuts down the battery if it gets too hot.

2. **Cell Balancing:** A lithium battery is made of many small "cells." The BMS ensures all cells are at the same voltage level. If one cell is weaker, the BMS manages it so it doesn't damage the whole pack.
3. **Communication:** Modern BMS systems communicate with the hybrid inverter, telling it the exact "State of Charge (SoC)," temperature, and health.

3.5 ADVANCED BATTERIES: FLOW BATTERIES

This is a large-scale, industrial-grade technology now becoming available for commercial use. The most common is the **Vanadium Redox Flow Battery (VRFB)**.

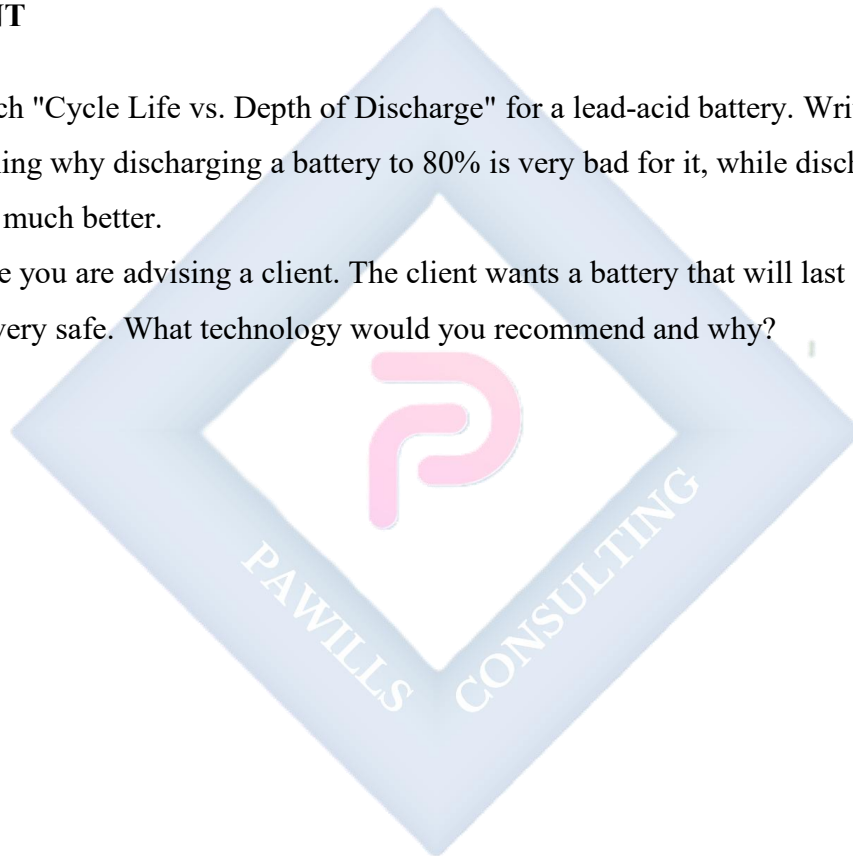
- **How it works:** It is completely different from other batteries. It's like a rechargeable fuel cell.
 - It has two large tanks of liquid electrolyte (one positive, one negative).
 - To charge or discharge, these liquids are *pumped* through a central "stack" where the chemical reaction happens.
- **Key Concept: Decoupled Power and Energy.**
 - The **Power (kW)** is determined by the size of the *stack*.
 - The **Energy (kWh)** is determined by the size of the *tanks*.
 - *Need more storage? Just install bigger tanks!*
- **Pros:**
 - **Extreme Cycle Life:** 10,000+ cycles. Lasts 20+ years.
 - **100% Depth of Discharge:** Can be fully discharged to 0% with no damage.
 - **Very Safe:** The liquids are not flammable.
 - **Scalable:** Perfect for massive, utility-scale storage (e.g., powering a whole factory).
- **Cons:**
 - **Very Expensive:** High upfront cost.
 - **Very Large & Complex:** Requires pumps and plumbing.
 - **Low Energy Density:** Takes up a lot of space.

EVALUATION

1. What is the maximum recommended Depth of Discharge (DoD) for a lead-acid battery?
2. What is the specific type of Lithium-ion battery most recommended for solar storage, and why?
3. List three main functions of a Battery Management System (BMS).
4. What is the unique feature of a flow battery that separates its Power (kW) from its Energy (kWh)?
5. Compare Lead-Acid and Lithium-ion in terms of Cycle Life, DoD, and Cost.

ASSIGNMENT

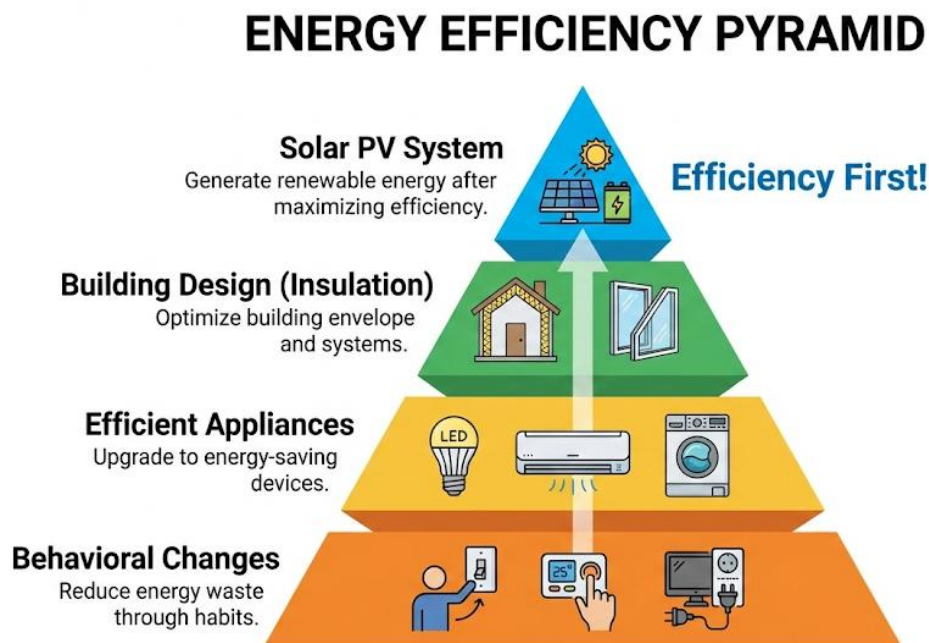
1. Research "Cycle Life vs. Depth of Discharge" for a lead-acid battery. Write a paragraph explaining why discharging a battery to 80% is very bad for it, while discharging it to 30% is much better.
2. Imagine you are advising a client. The client wants a battery that will last for 15 years and is very safe. What technology would you recommend and why?



WEEK 4: ENERGY EFFICIENCY IN SOLAR SYSTEMS

4.1 INTRODUCTION

This is one of the most important but overlooked topics. Many clients believe solar is a magic bullet. But the cheapest and greenest Watt (W) is the "**Negawatt**"—a Watt of energy that you *don't* use. **Core Principle:** It is always cheaper to *save* energy than to *produce* it.



4.2 THE IMPACT OF EFFICIENCY ON SYSTEM DESIGN

Look back at our Week 2 calculation. Our system was 1.1kWp.

- What if the client used old 100W incandescent bulbs instead of 10W LEDs?
 - Load: $10 \times 100\text{W} \times 8\text{h} = 8,000 \text{ Wh}$ (for lights *alone*).
 - The total load would *double*, meaning the client would need to buy *twice* the number of panels and *twice* the number of batteries.
- **Conclusion:** By investing ₦20,000 in LED bulbs, the client can save ₦1,000,000 on the cost of their solar system.

4.3 ENERGY CONSERVATION MEASURES

An installer's job begins with an **Energy Audit** to advise the client on conservation *before* designing the system.

1. **Lighting:**

- **Incandescent Bulbs:** 90% heat, 10% light. Highly inefficient.
- **CFL (Compact Fluorescent) Bulbs:** "Energy savers." Better, but contain mercury.
- **LED (Light Emitting Diode):** The *gold standard*. 90% light, 10% heat. Uses 1/10th the power of an incandescent.

2. **High-Efficiency Appliances:**

- Look for appliances with an **Energy Star** rating or a high efficiency label.
- **Refrigerators:** Modern inverter-compressor fridges use a fraction of the power of old ones.
- **Fans:** DC-powered fans are becoming popular, using 70% less power than AC fans.
- **Water Heaters:** A major "energy vampire." Advise clients to use solar water heaters (thermal) instead of electric ones.

3. **Phantom Loads (Vampire Loads):**

- These are devices that use electricity even when they are "off" (e.g., TVs on standby, phone chargers left plugged in, DSTV decoders).
- **Solution:** Advise clients to use power strips to turn these devices off completely.

4. **Behavioral Changes:**

- Train the client to be "energy-aware."
- Turn off lights when leaving a room.
- Use high-power appliances (like pumping machines, irons) in the *middle of the day* when the sun is at its peak, so the power comes directly from the panels, not the battery.

5. **Building Efficiency (Passive Design):**

- **Insulation:** Good ceiling insulation (like rock wool) stops heat from entering the building, reducing the need for AC.
- **Cross-Ventilation:** Designing windows for airflow.

- **Shading:** Planting trees or building awnings to stop the sun from hitting windows directly.

EVALUATION

1. What is a "Negawatt"?
2. Explain what a "Phantom Load" is and give two examples.
3. Why is it "cheaper to save energy than to produce it"?
4. What is the most efficient type of lighting available today?
5. What is the best time of day to use a high-power appliance in a solar-powered home, and why?

ASSIGNMENT

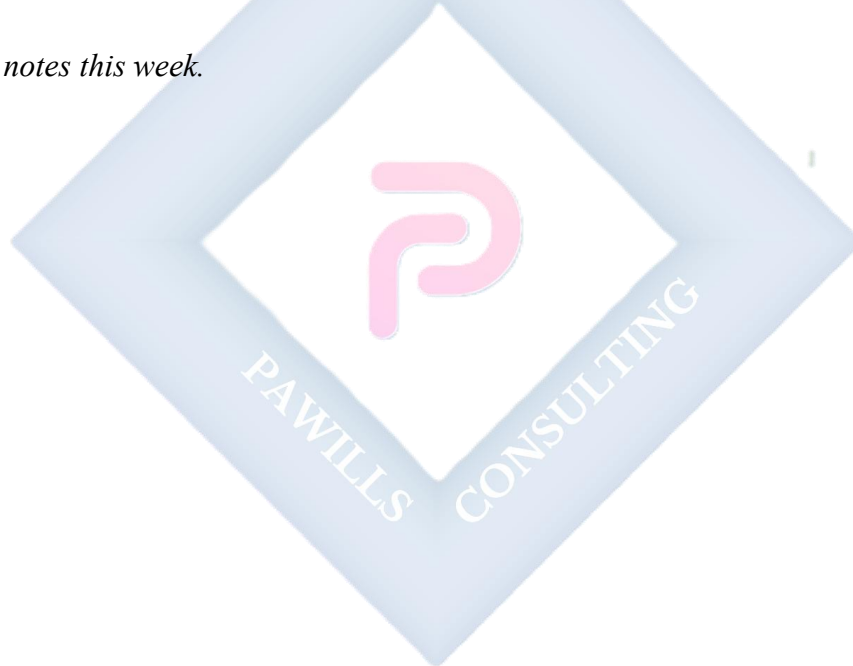
1. Conduct a simple "Energy Audit" of your own home or a room in your house.
2. Make a list of all electrical appliances.
3. Identify three (3) potential energy-saving opportunities (e.g., "replace incandescent bulb with LED," "unplug phone charger," "fix leaking fridge seal").
4. Write a short report on your findings.

WEEK 5: MIDTERM TEST

This week is for the midterm test, which will cover all topics from Weeks 1-4.

- **Week 1:** Review of SS1 basics, components (Panel, Controller, Battery, Inverter), and basic wiring (Series/Parallel).
- **Week 2:** Advanced system types (Off-grid, Grid-tied, Hybrid), Net-Metering, and the 4-step process for system sizing (Load, Inverter, Battery, Array).
- **Week 3:** Advanced battery technologies, comparison of Lead-Acid vs. Lithium-ion (LFP), functions of a BMS, and the concept of Flow Batteries.
- **Week 4:** Energy efficiency, "Negawatts," load audits, phantom loads, and conservation measures.

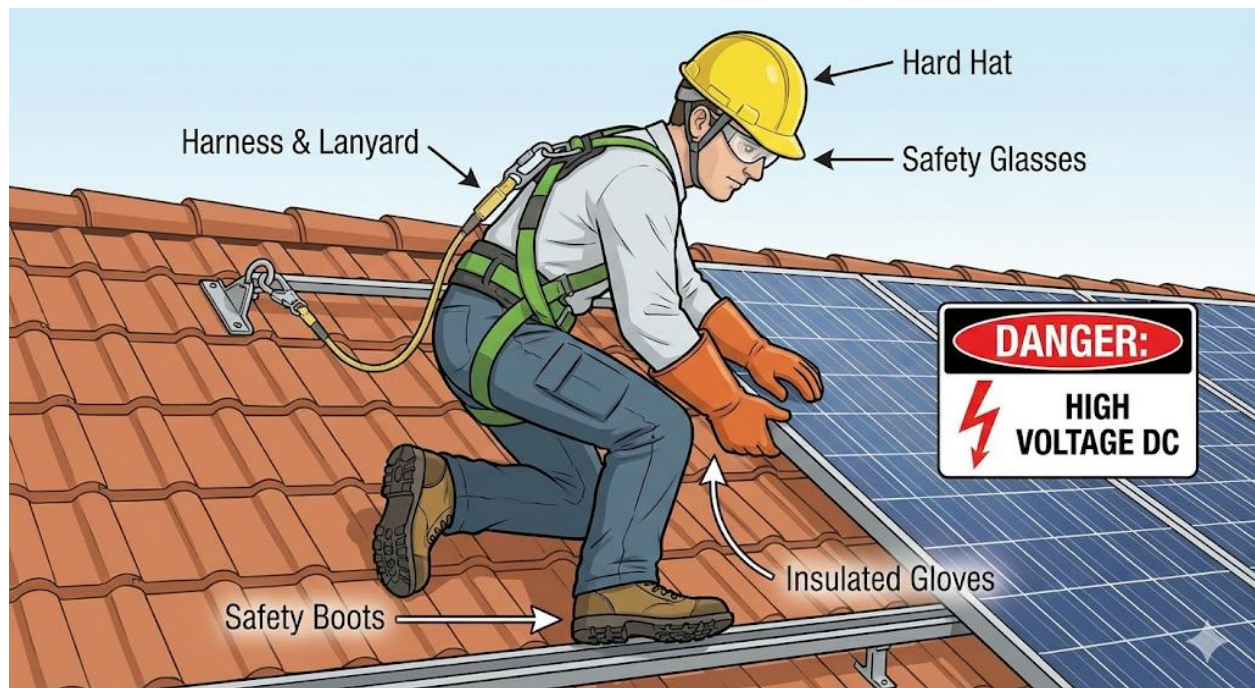
No new lesson notes this week.



WEEK 6: SAFETY IN PV INSTALLATION

6.1 INTRODUCTION

Solar installation is *not* a low-risk job. It involves working with live electricity (high-voltage DC) and working at heights. Safety is the #1 priority. A dead installer cannot fix a system.



6.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

This is the *minimum* equipment every installer must wear on a job site.

1. Fall Protection:

- **Safety Harness & Lanyard:** The most critical PPE when working on a roof. It must be securely anchored.

2. Head Protection:

- **Hard Hat:** Protects from falling tools or bumping your head.

3. Eye Protection:

- **Safety Glasses:** Protects from flying debris during drilling and from the bright glare of the sun (UV rated).

4. Hand Protection:

- **Insulated Gloves (Class 0):** These are special, rubber-insulated gloves rated to protect against electric shock (up to 1000V). **This is the most important electrical PPE.**
- **Work Gloves:** For general carrying and mounting, to protect from sharp panel edges.

5. Foot Protection:

- **Safety Boots:** Must have steel toes (to protect from falling panels) and **non-slip rubber soles** (to grip the roof).

6. Body Protection:

- Long-sleeved shirts (for sun protection).

6.3 SITE HAZARDS AND MITIGATION

A. ELECTRICAL HAZARDS

- **DC Arc Flash:** High-voltage DC (from a panel array) is *more dangerous* than AC. If a live DC wire is disconnected, the electricity can jump through the air, creating an "arc flash"—a 19,000°C explosion of plasma.
 - **Mitigation:** NEVER disconnect a live DC wire. Always use a DC disconnect switch. Work on one panel at a time.
- **Electric Shock:** An array of 10 panels can produce 400-600V DC, which is lethal.
 - **Mitigation:** Wear insulated gloves. Use insulated tools. De-energize the system. Follow **LOTO (Lockout/Tagout)**.
- **LOTO (Lockout/Tagout):** A safety procedure. Before working on a system, you "lock" the circuit breakers in the OFF position and put a "tag" with your name on it. This ensures no one can accidentally turn the power on while you are working.

B. PHYSICAL HAZARDS

- **Working at Heights (Falls):** This is the most common cause of death for installers.
 - **Mitigation:** Use a Fall Arrest System (harness). Ensure ladders are on stable ground and set at the correct angle (4:1 rule).

- **Lifting:** Solar panels are large and heavy.
 - **Mitigation:** Use proper lifting technique (lift with the legs, not the back). Use two people to carry a panel.
- **Slips and Trips:** Roofs can be slippery (especially in the morning dew). Tools left lying around are a trip hazard.
 - **Mitigation:** Wear non-slip boots. Maintain good "housekeeping" on the site.

C. ENVIRONMENTAL HAZARDS

- **Heatstroke / Sunburn:** Nigeria is hot. Working on a roof reflects sunlight and heat.
 - **Mitigation:** Drink plenty of water (not minerals/soda). Take regular breaks in the shade. Wear a wide-brim hat and use sunscreen.

EVALUATION

1. What does "PPE" stand for? List five (5) essential pieces of PPE for a solar installer.
2. What is a "DC Arc Flash," and why is it more dangerous than an AC arc?
3. Explain the "LOTO" procedure and its purpose.
4. What is the single most common cause of death or serious injury for a solar installer?
5. What is the most important piece of PPE for protecting against electric shock?

ASSIGNMENT

1. Draw a poster for a worksite titled "SOLAR SAFETY IS #1."
2. The poster must illustrate at least three different hazards (e.g., fall from height, electric shock, heavy lifting) and the correct PPE or mitigation for each.

WEEK 7: MIDTERM BREAK

This week is the midterm break. No new lesson notes.



WEEK 8: NIGERIAN & INTERNATIONAL STANDARDS

8.1 INTRODUCTION

A solar installation is not just a personal project; it's an electrical installation that must be safe, reliable, and compliant with national and international laws. Using sub-standard products or installation methods is dangerous and illegal.



8.2 NIGERIAN STANDARDS & BODIES

1. SON (Standards Organisation of Nigeria):

- **Function:** SON is responsible for setting the standards for *products* sold in Nigeria.
- **Relevance:** They set the **NIS (Nigerian Industrial Standard)** for PV panels, inverters, and batteries.
- **SONCAP:** This is the "Standards Organisation of Nigeria Conformity Assessment Programme." All imported solar products *must* go through this program to ensure they are not sub-standard *before* they can be sold in Nigeria.

- **As an installer:** You must check products for the SONCAP logo or NIS mark to ensure you are not using "fakes."
2. **NERC (Nigerian Electricity Regulatory Commission):**
- **Function:** NERC is the government body that regulates the *entire Nigerian electricity industry*.
 - **Relevance:** They are not concerned with your panels, but with *how your system affects the grid*.
 - **Key Regulations:**
 - **Mini-Grid Regulation:** NERC created the framework that allows private companies to build and operate "mini-grids" (large-scale solar systems) to power entire communities.
 - **Net-Metering Regulation:** NERC developed (and is still refining) the rules for grid-tied systems. This regulation mandates how DISCOs (like Ikeja Electric) must buy back excess power from customers.
 - **Installer Certification:** NERC has a program for certifying solar installers to ensure they are qualified.
3. **COREN (Council for the Regulation of Engineering in Nigeria):**
- **Function:** COREN regulates the *practice of engineering* in all its forms.
 - **Relevance:** Large-scale solar installations must be designed and signed off by a COREN-registered electrical engineer.

8.3 INTERNATIONAL STANDARDS (THE GLOBAL BENCHMARK)

These standards are used by SON to create the Nigerian (NIS) standards. They are the global test of quality.

1. **IEC (International Electrotechnical Commission):**
- This is the *most important* standards body for all electrical and electronic technology. When you buy a panel, you *must* check that it is "IEC Certified."
 - **IEC 61215 (Crystalline Panels):** This standard tests the *design and performance* of a panel. It's a "torture test" that includes:

- **Thermal Cycling:** Freezing the panel (-40°C) and then heating it (+85°C) 200 times.
 - **Hail Impact Test:** Firing balls of ice at the panel at high speed.
 - **Humidity-Freeze:** Soaking the panel in 85% humidity and then freezing it.
 - **IEC 61730 (PV Module Safety):** This standard tests *safety*. It ensures the panel won't give you a shock, that the glass is strong, and that the junction box is fire-resistant.
 - **IEC 62109 (Inverter Safety):** The equivalent safety standard for inverters.
2. **ISO (International Organization for Standardization):**
- **ISO 9001:** This doesn't certify the *product*, but the *manufacturer*. It certifies that the company has a good **Quality Management System** and that its factory processes are consistent and high-quality.

EVALUATION

1. What is the role of SON in the Nigerian solar industry?
2. What is SONCAP?
3. What is the role of NERC? How is it different from SON?
4. What is the most important *international* standards body for solar panels?
5. What is the difference between IEC 61215 and IEC 61730?

ASSIGNMENT

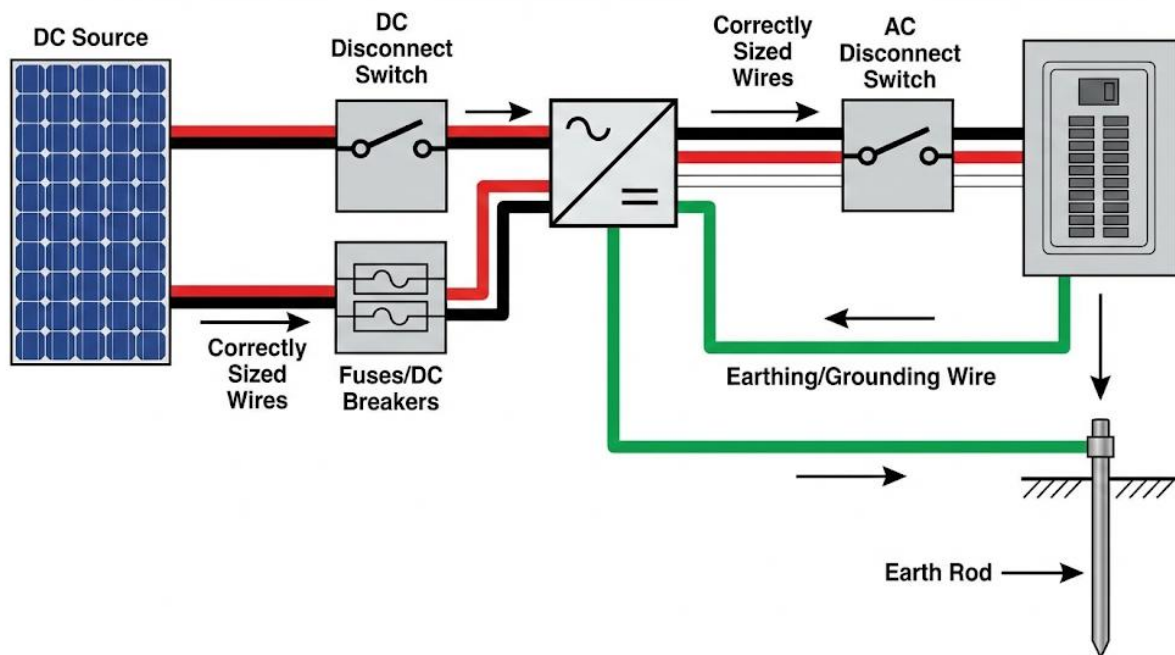
1. Find a website selling solar panels in Nigeria (e.g., Jumia, or any solar installer's site).
2. Look at the "specifications" for a panel. Try to find the "Certifications" section.
3. Write down all the certifications you can find listed (e.g., "IEC 61215," "ISO 9001," "SONCAP").

WEEK 9: ELECTRICAL CODES & REGULATIONS

9.1 INTRODUCTION

Standards (Week 8) tell you the *quality* of a product. **Codes** tell you *how to install it*. An electrical code is the rulebook for all electrical installations. Its only purpose is **safety**—to prevent fires and electric shock.

In Nigeria, the "Nigerian Wiring Code" is often based on international codes, primarily the British standard **BS 7671** or the **IEC 60364** (Low-voltage electrical installations).



9.2 KEY CODE REQUIREMENTS FOR PV SYSTEMS

1. Wire Sizing (Cable Ampacity):

- This is one of the biggest causes of solar fires. If a wire is too small for the current (Amps) passing through it, it will overheat and melt.
- **Ampacity:** The maximum current a wire can safely carry.
- **Voltage Drop:** Using a wire that is too small over a long distance (e.g., from the roof to the inverter) will cause a "voltage drop," wasting energy.

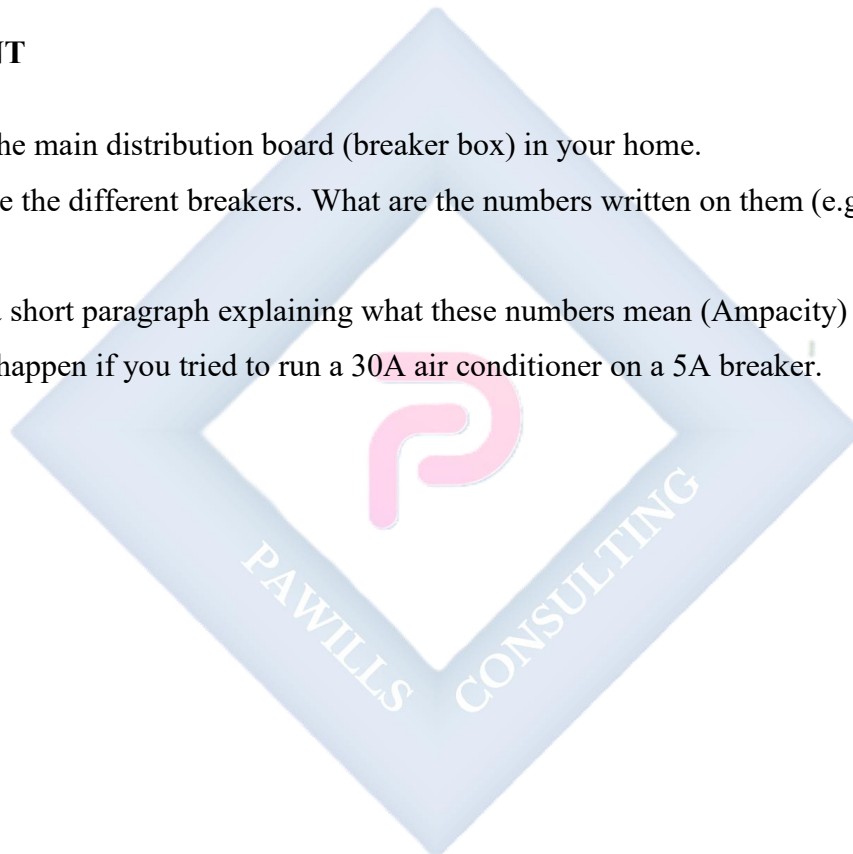
- **Code:** Wires must be sized to handle the "Short Circuit Current (I_{sc})" of the array, plus a safety factor. They must also be **UV-resistant** (if outdoors).
2. **Overcurrent Protection (Fuses & Breakers):**
- **Function:** To protect the *wires* from overheating. A fuse or breaker is a "sacrificial" device that blows (fused) or trips (breaker) if the current gets too high.
 - **Code:** You *must* have:
 - **DC Fuses/Breakers:** Between the panels and the charge controller.
 - **DC Fuses/Breakers:** Between the controller and the battery.
 - **AC Breakers:** Between the inverter and the AC loads.
 - **Note:** DC breakers are *different* and more expensive than AC breakers.
3. **Disconnects (Isolators):**
- **Function:** These are "on/off" switches that allow an installer to safely "isolate" components for maintenance.
 - **Code:** A lockable DC disconnect (isolator) *must* be installed between the panels and the inverter. An AC disconnect *must* be installed between the inverter and the building's electrical panel.
4. **Earthing (Grounding):**
- This is a critical safety component. It provides a safe path for stray electricity to go into the "earth" (ground) instead of through a person.
 - **Code:**
 - **Equipment Grounding:** All metal, non-current-carrying parts *must* be grounded. This includes the *frames* of the solar panels, the metal mounting racks, and the chassis (metal box) of the inverter. This is done by connecting them all with a green/yellow earthing wire to a copper **earthing rod** (driven deep into the ground).
 - **System Grounding:** (More complex) Involves grounding one of the current-carrying conductors (usually the negative).
5. **Wire Management (Conduit):**
- **Code:** All outdoor PV wiring must be protected from physical damage and sunlight. This is done by running the wires inside a **conduit** (either metal or UV-resistant PVC pipe). Wires cannot just be left dangling.

EVALUATION

1. What is the primary purpose of an electrical code?
2. Why is wire sizing (ampacity) so important in a PV installation?
3. What is the difference between an "Overcurrent Protection" device (like a fuse) and a "Disconnect" (like an isolator)?
4. What two things *must* be earthed (grounded) in a PV installation?
5. What is a "conduit," and why is it required by code?

ASSIGNMENT

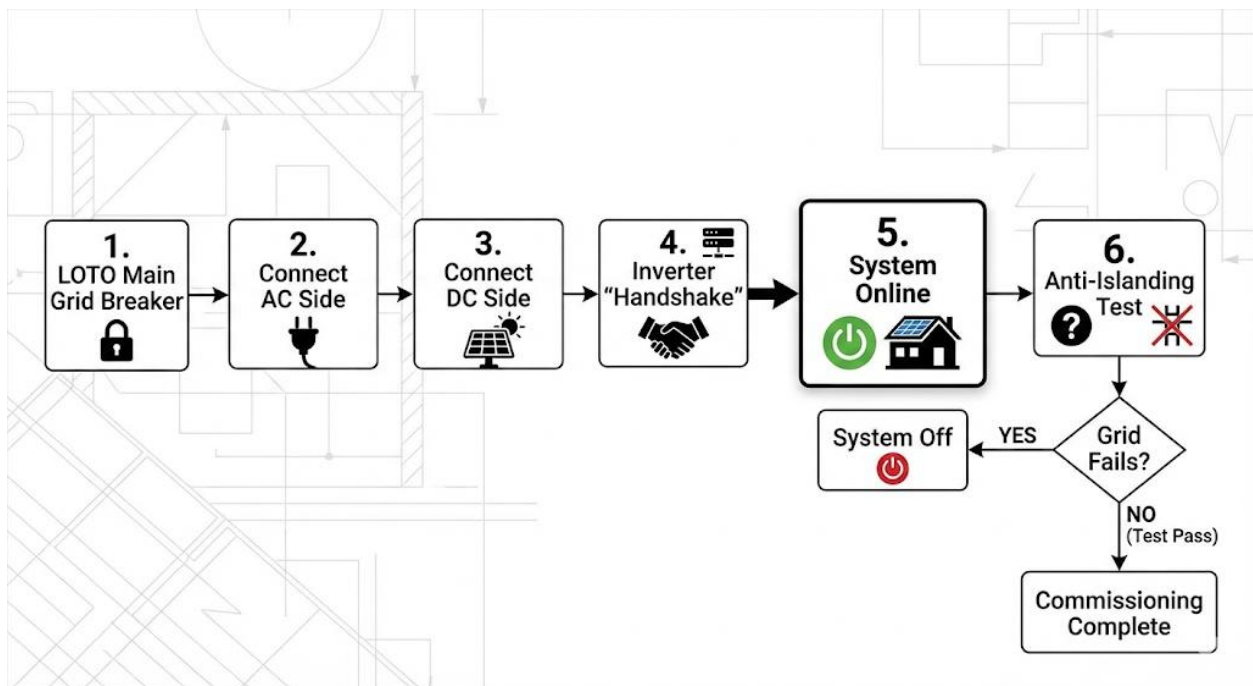
1. Go to the main distribution board (breaker box) in your home.
2. Observe the different breakers. What are the numbers written on them (e.g., 5A, 15A, 30A)?
3. Write a short paragraph explaining what these numbers mean (Ampacity) and what would happen if you tried to run a 30A air conditioner on a 5A breaker.



WEEK 10: PRACTICAL: GRID-TIED SOLAR SETUP (DEMONSTRATION)

!! SAFETY WARNING !! THIS IS A LESSON FOR OBSERVATION AND

DEMONSTRATION ONLY. A grid-tied system is connected to the NEPA/PHCN grid. It involves lethal AC voltages (240V) and lethal DC voltages (up to 600V). It can also "back-feed" the grid, potentially electrocuting utility workers. **STUDENTS MUST NOT ATTEMPT THIS.** This practical must only be demonstrated by a qualified, licensed instructor.



10.1 OBJECTIVE

To demonstrate the components, connection sequence, and safety features (especially anti-islanding) of a grid-tied solar system.

10.2 COMPONENTS FOR DEMONSTRATION

- PV Panels (2 x 300W panels)
- **Grid-Tied Inverter** (e.g., 600W Micro-inverter or String Inverter)
- DC Disconnect Switch
- AC Disconnect Switch

- Dedicated AC Breaker
- Main Distribution Board (connected to grid/mains power)
- Bi-Directional Meter (if available) or standard "spinning disc" meter.
- PPE (Instructor must wear insulated gloves, safety glasses)

10.3 DEMONSTRATION STEPS

STEP 1: SAFETY & LOTO

- Instructor shows the LOTO (Lockout/Tagout) on the main AC breaker. The entire demonstration area is de-energized.
- Instructor puts on all PPE.

STEP 2: MECHANICAL MOUNTING (Review)

- Instructor briefly reviews mounting panels on a demo roof frame, ensuring they are secure and earthed (a ground wire is connected to the panel frame).

STEP 3: DC WIRING

- Instructor connects the two panels *in series* (+ to -).
- The final array output (+ and -) is run in a conduit to the **DC Disconnect Switch**.
- The output of the DC Disconnect is connected to the **DC Input** of the grid-tied inverter.
- **Instructor's Note:** "The DC side is now complete. We *never* close this switch until the AC side is 100% complete and energized."

STEP 4: AC WIRING

- Instructor connects the **AC Output** of the inverter to the **AC Disconnect Switch**.
- The output of the AC Disconnect is connected to a *new, dedicated AC Breaker* inside the main distribution board.
- **Instructor's Note:** "This is the grid connection. This inverter is now physically connected to NEPA. It is an *appliance*, just like an AC unit."

STEP 5: COMMISSIONING (THE "HANDSHAKE")

- This is the most important step.
- **Instructor:** "The inverter is still *off*. It has no 'on' switch. It is a smart device. It will not turn on *unless* it detects a stable grid. This is its primary safety feature."
- **Sequence:**
 1. Check all wiring. S. Remove LOTO and turn ON the **Main Grid Breaker**.
(Power flows to the inverter).
 2. Turn ON the **AC Disconnect** and the **new AC Breaker**.
 3. **Instructor:** "The inverter's 'brain' is now on. It is *sensing* the grid. It is checking the grid's voltage (230V?) and frequency (50Hz?). This 'handshake' can take 30-60 seconds."
 4. Turn ON the **DC Disconnect Switch**. (DC power now flows from the panels).
 5. After a few moments, the inverter's screen/light will turn GREEN. It is now "on-line" and producing power. We can see the Watts being produced on its display.
 6. **Instructor:** "If we look at the utility meter, we can see it has slowed down or, if we are producing *more* power than we are using, it will start to spin *backwards* (Net Metering)."

STEP 6: SAFETY TEST (ANTI-ISLANDING)

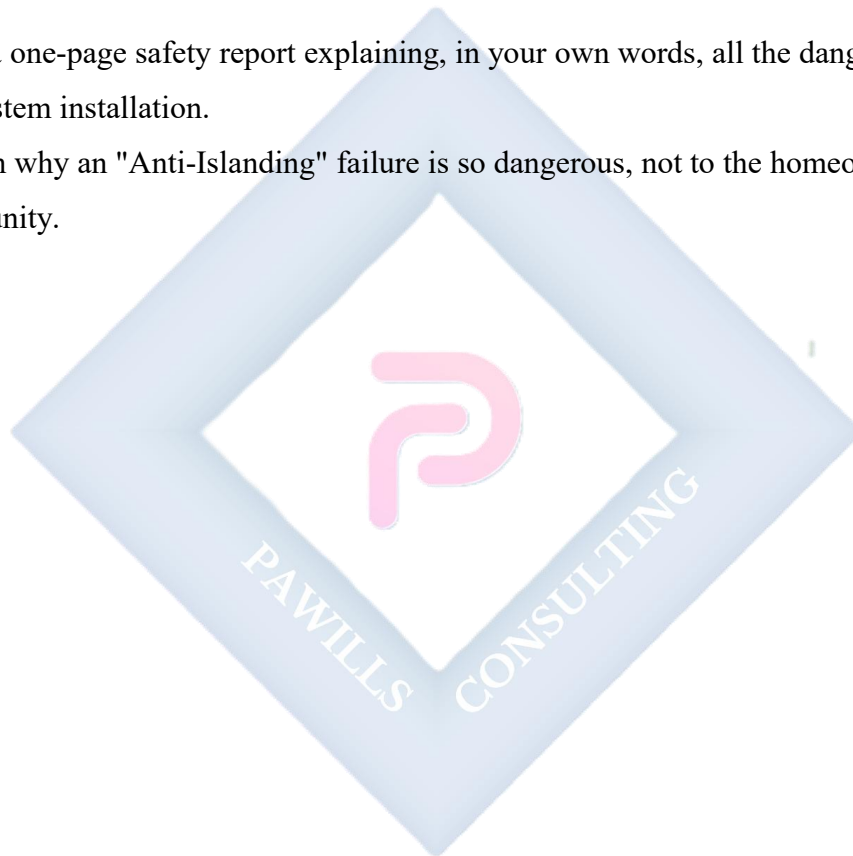
- **Instructor:** "What happens if NEPA goes off? We must be sure our inverter *stops* sending power to the grid, or a NEPA worker fixing the line could be killed. This is called 'Anti-Islanding'."
- **Test:** The instructor suddenly switches **OFF** the **Main Grid Breaker** (simulating a power cut).
- **Observation:** The grid-tied inverter must shut down *instantly* (within milliseconds). The green light will turn red or off.
- **Conclusion:** The inverter is safe. It only works when the grid is present.

EVALUATION

1. What is the "Anti-Islanding" feature? Why is it the most important safety feature of a grid-tied inverter?
2. What is the correct "power-on" (commissioning) sequence? (DC first or AC first?)
3. Why is a DC disconnect switch necessary?
4. Why must a grid-tied inverter be connected to a *dedicated* AC breaker?
5. What is the "handshake"? What two things is the inverter checking for?

ASSIGNMENT

1. Write a one-page safety report explaining, in your own words, all the dangers of a grid-tied system installation.
2. Explain why an "Anti-Islanding" failure is so dangerous, not to the homeowner, but to the community.



WEEKS 11–13: REVISION & EXAMS

11.1 SUMMARY OF TERM 1

This term, we made the leap from basic SS1 concepts to professional SS2-level design and compliance.

- **We reviewed** the core components and wiring of off-grid systems.
- **We learned to design** advanced **Hybrid** and **Grid-Tied** systems, including the 4-step calculation process (Load, Inverter, Battery, Array).
- **We explored** modern **Energy Storage**, comparing Lead-Acid to **Lithium-ion (LFP)** and **Flow Batteries**, and understood the critical role of a **BMS**.
- **We prioritized Energy Efficiency**, learning that "Negawatts" and conservation are the first steps in system design.
- **We mastered Safety (PPE, LOTO)**, learning about the extreme dangers of DC Arc Flash and working at heights.
- **We understood** the regulatory landscape, from **Nigerian (SON, NERC)** standards to **International (IEC)** certifications.
- **We learned** the fundamentals of **Electrical Codes**, including wire sizing, overcurrent protection, and earthing.
- **We observed** the commissioning and safety checks (especially **Anti-Islanding**) of a **Grid-Tied** system.

11.2 FINAL EXAM PREPARATION

- Review all calculations from Week 2. Be ready to size a simple system.
- Be able to compare and contrast Lead-Acid and Lithium-ion.
- Be able to list at least 5 essential PPE items and 3 major site hazards.
- Be able to explain the difference between SON, NERC, and the IEC.
- Be able to define "Anti-Islanding" and "Net Metering."

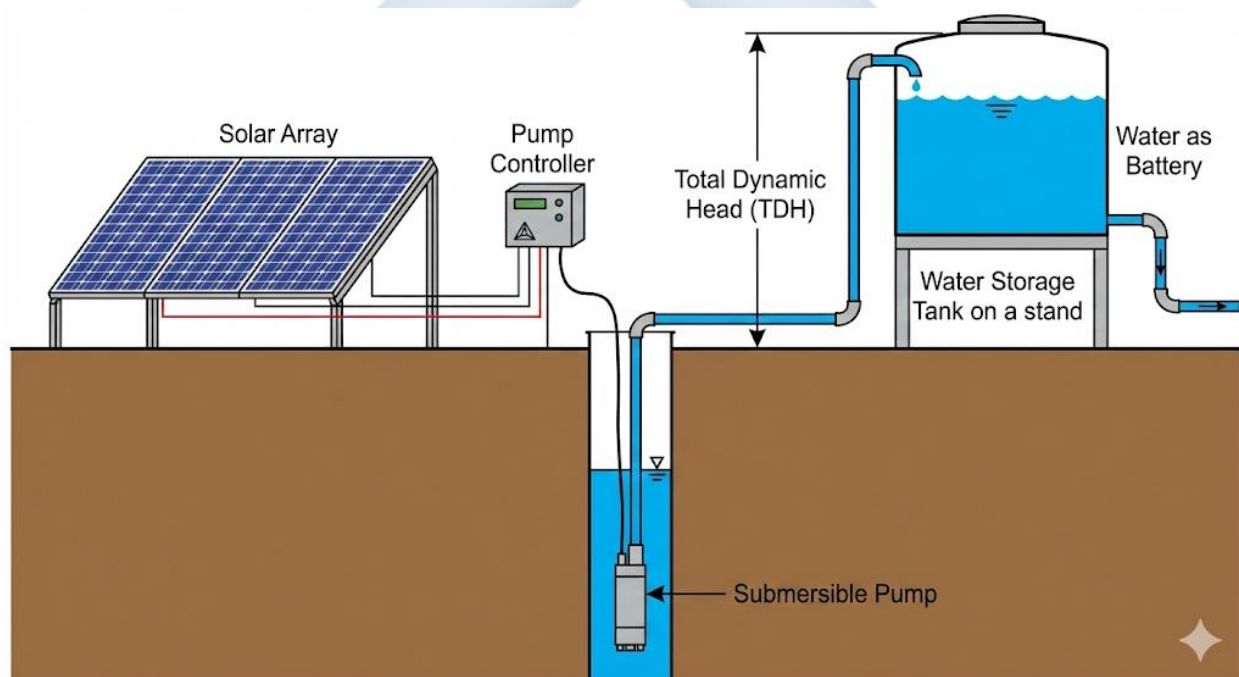
SS2 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

SECOND TERM

WEEK 1: SOLAR WATER PUMPING SYSTEMS

1.1 INTRODUCTION

This week, we explore one of the most transformative applications of solar power, especially in Nigeria and across Africa: solar water pumping. This technology provides a reliable, cost-effective, and off-grid solution for accessing water, which is fundamental for agriculture, livestock, and rural community life. Unlike systems for homes, these are specialized to power a single, heavy-duty load: a motor.



1.2 DESIGN AND SYSTEM TOPOLOGY

A solar water pumping system is elegantly simple and robust. Its design philosophy is different from a battery-based home system.

Core Components:

1. PV Array (Solar Panels):

- **Function:** Provides the raw DC electricity. The size of the array (in Watts) is determined by the power required by the pump.
- **Configuration:** The panels are wired in series/parallel to match the specific voltage (V) and current (A) requirements of the pump controller.

2. Solar Pump Controller (The "Brain"):

- This is **not** a charge controller. It is a highly specialized inverter called a **Variable Frequency Drive (VFD)** or a **DC-DC Converter**.
- **Function 1: Power Conversion:**
 - **For DC Pumps:** It acts as a smart DC-DC converter, optimizing the panel's voltage for the DC motor.
 - **For AC Pumps:** (More common for larger systems) It converts the high-voltage DC from the panels into high-quality 3-phase AC power for the motor.
- **Function 2: MPPT (Maximum Power Point Tracking):** It constantly adjusts the electrical load to find the "sweet spot" of the solar array, ensuring the pump gets the maximum possible power at any given moment of sunlight.
- **Function 3: Protection (The Most Critical Job):**
 - **Dry-Run Protection:** It receives signals from well-probes. If the water level in the borehole drops too low, the controller *shuts off the pump* to prevent it from burning out.
 - **Motor Protection:** Protects the motor from over/under-voltage, over-current, and overheating.
- **Function 4: Soft Start:** It gradually "ramps up" the pump's speed in the morning as the sun rises, preventing the massive "inrush current" (power surge) that would be damaging.

3. The Pump & Motor:

- **Submersible Pump:** The most common type. The entire unit (pump and sealed motor) is lowered deep into a borehole or well, fully submerged in water. This is excellent for deep wells and high "lift."

- **Surface Pump:** The pump and motor are located on the ground. It *pulls* water from a shallow source (river, lake, shallow well) and pushes it.
4. **Storage: The Water Tank (The "Battery"):**
- **Key Concept:** In over 90% of solar pumping systems, **there are no batteries.** Batteries are the most expensive, complex, and short-lived part of a solar system.
 - **The "Battery" is Water:** The design uses the sun's energy to pump water all day long into a large, elevated storage tank (e.g., a 10,000-litre tank on a 5-meter stand). This stored *water* is then used 24/7 via gravity. This is far cheaper and more reliable than storing electricity in chemical batteries.
5. **Ancillary Components:**
- **Well Probes:** Sensors that detect the water level in the borehole.
 - **Float Switch:** A switch placed in the *storage tank*. When the tank is full, it floats to the top and sends a signal to the controller to *stop the pump* (preventing overflow).

1.3 SYSTEM SIZING AND DESIGN

Sizing a pump is a critical calculation based on two factors: **Flow** and **Head**.

1. **Flow (Volume):**

- How much water does the client need per day?
- This is measured in **Cubic Meters per day (m³/day)** or **Litres per day**. (1,000 litres = 1 m³).
- *Example:* A small community needs 20,000 litres/day. A farm needs 50,000 litres/day for irrigation.

2. **Total Dynamic Head (TDH):**

- This is the total "pressure" the pump must fight against. It is measured in meters (m).
- **TDH = Static Head + Friction Head**
- **Static Head:** The total *vertical* distance the water must be lifted.
 - *Formula:* (Depth of water in the well) + (Vertical height from ground to the tank).

- *Example:* Water is 70m deep in the borehole. The tank is on a 5m stand.

$$\text{Static Head} = 70\text{m} + 5\text{m} = 75\text{m}.$$

- **Friction Head:** The "extra" pressure caused by water rubbing against the inside of the pipes. The longer the pipe, the more friction. We add a safety factor (e.g., 10%) to account for this.

- *Example:* 10% of 75m = 7.5m.

- **Total TDH:** 75m + 7.5m = **82.5m**.

3. Sizing:

- The installer takes these two numbers (**e.g., 20 m³/day @ 82.5m TDH**) and consults a manufacturer's "pump curve" chart to select a pump model (e.g., a 1.5kW / 2HP pump).
- **PV Array Sizing:** The solar array must be oversized to account for losses (heat, dust, clouds).
- *Rule of Thumb:* Array Size = Pump Power x 1.5 (or more).
- *Example:* For a 1.5kW (1500W) pump, you would install: 1500W x 1.5 = **2,250Wp** (e.g., 5 x 450W panels).

1.4 APPLICATIONS IN AGRICULTURE

- **Irrigation:** This is the *killer application*. It allows farmers to practice **Dry Season Farming**, moving from 1 rain-fed harvest per year to 2 or 3 harvests.
- **Drip Irrigation:** Solar pumps are perfect for powering drip irrigation systems, delivering water and fertilizer directly to the roots of plants with minimal waste.
- **Livestock:** Provides a constant, reliable source of clean water for cattle, poultry, and fish ponds, increasing animal health and productivity.
- **Rural Water Supply:** Powers boreholes in villages, providing clean drinking water. This has a massive positive impact on health (reducing water-borne disease) and society (freeing women and children from the daily task of fetching water).

EVALUATION

1. Why do most solar water pumping systems **not** use batteries? What is used for storage instead?
2. What are the two (2) main functions of a solar pump controller?
3. Explain the purpose of a "dry-run protection" sensor.
4. What is "Total Dynamic Head (TDH)"? What two components make it up?
5. A farmer needs a pump. The water in his well is 50m deep, and his tank is 10m tall. What is the **Static Head**?

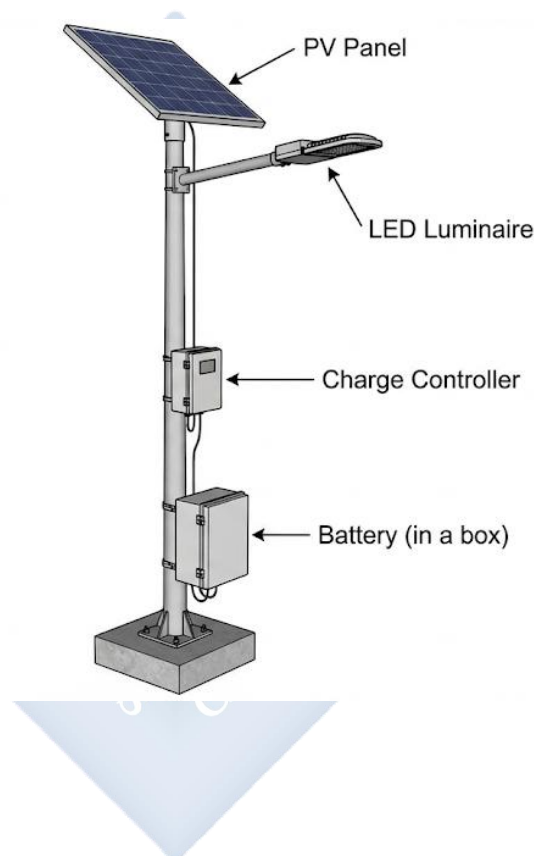
ASSIGNMENT

1. A community requires a 2kW solar pump. Using the "Rule of Thumb" ($\text{Array} = \text{Power} \times 1.5$), calculate the total size (in Wp) of the solar array needed.
2. If you decided to use 400W solar panels for the array in Q1, how many panels would you need?
3. Write a one-page report on the social and economic impact of "Dry Season Farming" in Northern Nigeria, enabled by solar irrigation.

WEEK 2: SOLAR STREET LIGHTING SYSTEMS

2.1 INTRODUCTION

Solar street lights are one of the most visible and common applications of PV technology. Each street light is a complete, self-contained **standalone (off-grid)** system. They are designed for automation, durability, and providing security and illumination in areas with no or unreliable grid power.



2.2 SYSTEM COMPONENTS

A solar street light is composed of five main parts, typically mounted on a single pole.

1. PV Panel:

- A single panel (e.g., 50W to 150W) sized to charge the battery in one day.
- It is mounted at the top of the pole with a bracket, angled to capture maximum sunlight.

2. Battery:

- This is the *heart* of the system. It is charged during the day and discharged *every single night*. This is a very high-cycle, demanding application.
- **Types:**
 - **Deep-Cycle Lead-Acid (AGM/Gel):** Cheaper, heavier. Fails quickly (1-2 years) because it cannot handle the nightly deep discharge. This is the main failure point of cheap systems.
 - **Lithium-ion (LiFePO4):** The modern standard. Lighter, 5-10 year lifespan, high DoD (80-90%).
- **Location:** Housed in a weatherproof, lockable box, either at the base of the pole (vulnerable to theft) or, more securely, near the top, just under the panel.

3. Solar Charge Controller (The "Brain"):

- A specialized controller that manages the battery and the light. It is **not** a standard MPPT controller.
- **Core Function: Dusk-to-Dawn Automation:**
 - **Daytime:** The controller senses a high voltage from the solar panel. This tells it "It is daytime." It then sends the power to the battery *to charge it* and keeps the light *OFF*.
 - **Nighttime:** When the sun sets, the panel's voltage drops to zero. The controller senses this ("It is nighttime"). It stops charging and switches the power flow, sending electricity *from* the battery *to* the light, turning it *ON*.
- **Other Features:**
 - **Dimming:** To conserve battery, it can be programmed (e.g., 100% brightness for the first 5 hours, then 50% for the rest of the night).
 - **Motion Sensor:** Many modern lights stay dim (e.g., 30%) and only go to 100% brightness when they detect a person or car passing, saving significant energy.

4. LED Luminaire (The Light Fixture):

- All modern solar street lights use **LEDs (Light Emitting Diodes)**.
- **Why LED?** They are extremely energy-efficient (more lumens per Watt), very durable, and have a long lifespan (50,000+ hours). They are perfect for battery-powered applications.

5. Pole and Mounting Hardware:

- A galvanized (anti-rust) steel pole, foundation cage, and the "arm" (bracket) that holds the light fixture.

2.3 INSTALLATION

1. **Civil Works (The Hardest Part):** A deep foundation (1-1.5m) must be dug. A rebar "cage" is built and placed in the hole, and the pole's base-plate is bolted to it. The hole is then filled with concrete.
2. **Assembly:** The panel, light fixture, and battery box are assembled on the pole *while it is on the ground*.
3. **Wiring:** The system is wired on the ground. It is a simple 12V or 24V DC system:
 - Panel wires -> "PV" terminals on the controller.
 - Battery wires -> "BATTERY" terminals on the controller.
 - Light wires -> "LOAD" terminals on the controller.
4. **Erection:** A crane (or a large team) is used to lift the entire assembled pole and place it onto the bolts on the concrete foundation.
5. **Commissioning:** The system is automatic. The final test is to cover the solar panel with a dark cloth or cardboard. This "tricks" the controller into thinking it's night, and the light should turn on after a few seconds.

2.4 MAINTENANCE

Solar street lights are "low-maintenance," not "no-maintenance."

1. **Panel Cleaning:** This is the **most important** task. The panels are flat and collect dust, bird droppings, and (in Lagos) salt spray. This "soiling" blocks sunlight and *starves* the battery.
 - **Schedule:** Must be cleaned 2-4 times per year (especially after Harmattan).
2. **Battery Replacement:** This is the main *corrective* maintenance.
 - **Lead-Acid:** Expect to replace it every 1-2 years.
 - **Lithium-ion:** Expect to replace it every 5-10 years.

- **Failure Point:** 90% of solar street light failures are due to *battery failure* (either premature death from using cheap lead-acid, or end-of-life).
3. **Visual Inspection:** Check for:
- **Theft/Vandalism:** (Batteries are a primary target).
 - **Damage:** (e.g., a car hitting the pole).
 - **Shading:** (e.g., a nearby tree has grown and is now blocking the panel).

EVALUATION

1. What are the five (5) main components of a solar street light system?
2. How does the charge controller "know" when to turn the light on?
3. Why are LEDs used for the light fixture instead of older bulb types?
4. What is the most important *preventive* maintenance task for a solar street light?
5. What is the single most common *failure point* (the component that usually breaks first)?

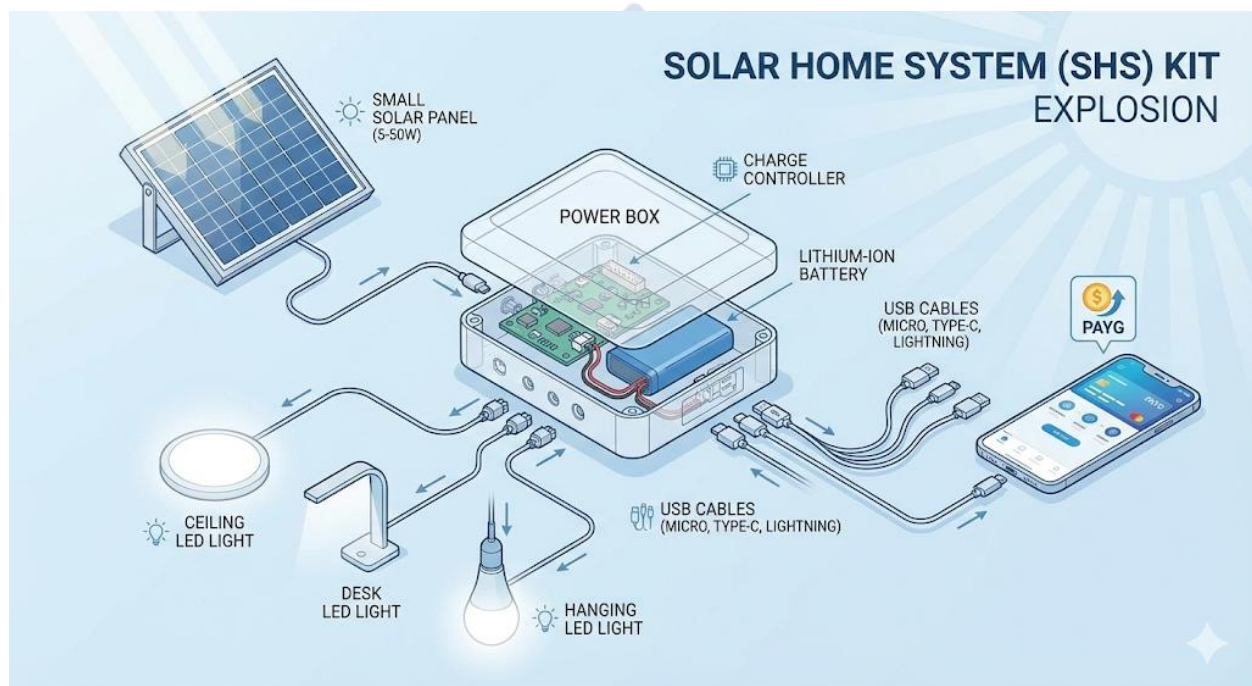
ASSIGNMENT

1. There are two main designs for solar street lights:
 - **Split Type:** The panel is separate from the light/battery (as described above).
 - **All-in-One:** The panel, battery, and LED are all integrated into a single, compact unit.
2. Research the Pros and Cons of "All-in-One" vs. "Split" solar street lights. Create a comparison table. (Hint: Think about theft, performance, and maintenance).

WEEK 3: SOLAR HOME SYSTEMS (SHS)

3.1 INTRODUCTION

A Solar Home System (SHS) is a small-scale, standalone (off-grid) kit designed to provide basic, essential power to a home, shop, or small business. These systems are the first step on the "energy ladder" and are critical for bringing light and communication to millions of rural and peri-urban households.



3.2 SYSTEM ARCHITECTURE (KITS)

An SHS is a complete "system in a box." It is designed for simplicity, not power.

1. **PV Panel:** A single, small panel (e.g., 20W, 50W, or 100W).
2. **Control/Battery Unit (The "Power Box"):**
 - This is the core of the system. It is a single, portable unit that contains:
 - A simple **charge controller** (usually PWM).
 - A small **battery** (e.g., 12V, 7Ah to 50Ah, either Lead-Acid or Lithium).

- A user-facing **interface** with switches, battery status indicators (LEDs), and multiple output ports.

3. Loads (The Appliances):

- **DC-Powered:** A key feature of SHS is that they primarily use **DC appliances**. This avoids the cost and energy loss (10-15%) of an inverter.
- **DC LED Lights:** The kit comes with 2-4 low-voltage (12V) DC LED bulbs with long cables.
- **DC USB Ports:** For charging mobile phones. This is often the #1 most desired feature.
- **DC Radio/Fan (Optional):** Some larger kits include a small DC fan or radio.

4. Inverter (Optional):

- Some larger, more advanced SHS kits (e.g., 150W+) may include a small, built-in inverter (e.g., 100W) to power *one* small AC appliance, like a small TV, laptop, or a barber's clipper.

3.3 PLUG-AND-PLAY DESIGN

The main feature of an SHS is that it is *user-installable*. It requires no technical skill, making it a "plug-and-play" retail product.

- **Plug-and-Play:** All components use simple, colour-coded, or unique connectors that cannot be plugged in the wrong way.
- **Example:** The wire from the panel has a specific plug that fits the "PV" socket. The light bulbs have long cables with small plugs that fit the "LIGHT" sockets.
- The user just mounts the panel on the roof (even with bricks) and connects the plugs.

3.4 PAY-AS-YOU-GO (PAYG) MODEL

This is the *business model* that unlocked the SHS market in Africa (pioneered by companies like M-Kopa, Lumos, and Azuri).

- **The Problem:** The upfront cost of an SHS (e.g., ₦50,000) is too high for a low-income rural family.

- **The Solution (PAYG):**

1. The customer pays a small deposit (e.g., ₦5,000) and gets the SHS kit.
 2. The kit is "locked" by the company.
 3. The customer makes small daily, weekly, or monthly payments (e.g., ₦100 per day) via mobile money (e.g., MTN, Airtel).
 4. When they pay, they receive a code to punch into the SHS keypad, "unlocking" it for the time they paid for.
 5. If they stop paying, the system locks itself, and the lights go out.
 6. After 1-3 years of payments, the system is fully paid for, and the company unlocks it permanently. The customer now *owns* it.
- **Why it works:** It turns a large capital expense into a small, predictable operating expense, often *cheaper* than their previous daily spending on kerosene, candles, and phone charging.

3.5 ADVANTAGES AND APPLICATIONS

- **Energy Access:** Brings light to homes for the first time.
- **Education:** Children can study after dark.
- **Health & Safety:** Replaces toxic, flammable kerosene lamps, reducing indoor air pollution and fire risk.
- **Economic:**
 - **Savings:** Saves money spent on kerosene and phone charging.
 - **Business:** A shop can stay open later; a "phone charging" business can be started; a barber can run clippers.

EVALUATION

1. What is a Solar Home System (SHS)?
2. Why do most SHS kits use DC-powered appliances instead of AC?
3. What is a "plug-and-play" design?
4. Explain the "PAY-AS-YOU-GO" (PAYG) business model. Why is it so important for SHS?

5. List three (3) social or economic benefits of SHS in a rural community.

ASSIGNMENT

1. In Nigeria, "Lumos" (with MTN) and "MTN" themselves have heavily marketed SHS kits.
2. Go online (or visit a local dealer) and find the specifications for one of these kits (e.g., the Lumos Prime or MTN SIM-enabled kits).
3. Write down: (a) The size of the solar panel (in Watts), (b) The number of lights it supports, (c) Whether it includes an inverter, (d) The payment plan (if you can find it).

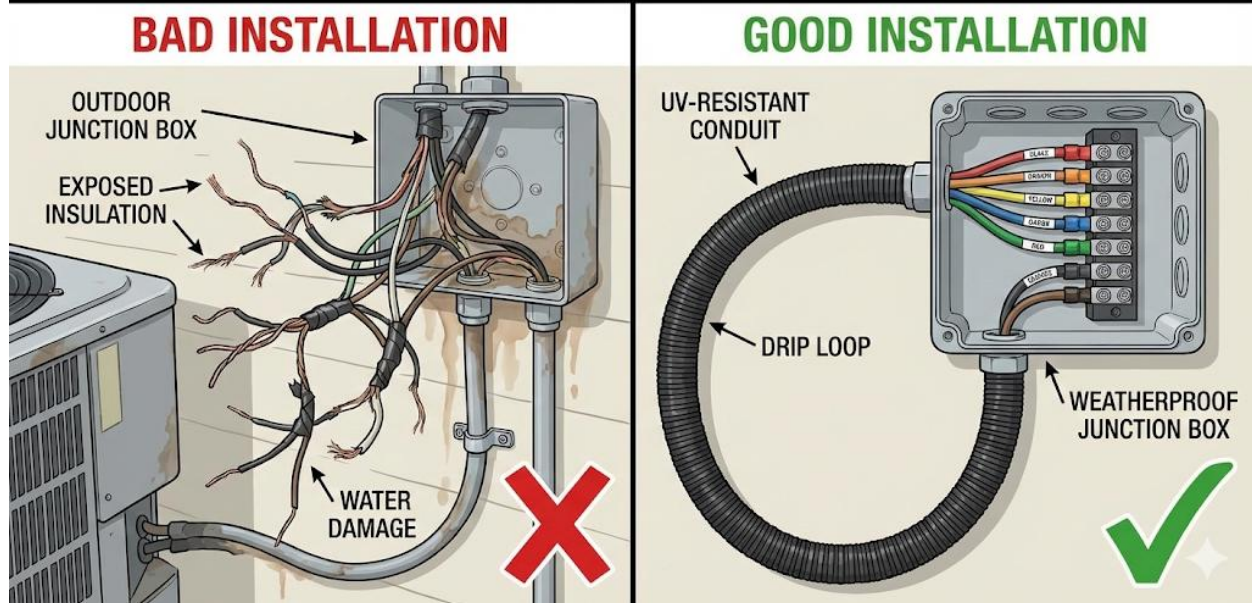


WEEK 4: ADVANCED WIRING TECHNIQUES

4.1 INTRODUCTION

An electrical installation is only as safe and reliable as its wiring. A solar installation is no different. Poor wiring is the **#1 cause of solar fires** and system failures. This week, we move beyond simple connections to the professional techniques required for a safe, durable, and code-compliant installation.

ADVANCED WIRING TECHNIQUES: INSTALLATION QUALITY



4.2 CABLE PROTECTION (CONDUITS AND TRUNKING)

Exposed wires are a sign of an amateur, unsafe installation. Wires must be mechanically protected from physical damage, rodents, and UV (sunlight).

1. Conduit (Pipes):

- **Function:** A pipe used to route and protect electrical wires.
- **Types:**
 - **PVC Conduit:** (White pipes) Common for indoor use. Cheap, but *not* UV-resistant. Becomes brittle and cracks in the sun.

- **UV-Resistant PVC Conduit:** (Grey or Black) Specially treated for outdoor use. This is the minimum standard for routing wires on a roof.
- **Flexible Conduit (Metallic/Non-Metallic):** A flexible "hose" used for "final-mile" connections, in tight bends, or to connect to equipment that vibrates.
- **EMT (Electrical Metallic Tubing) / Galvanized Steel:** (Silver metal pipes) Required for industrial/commercial jobs. Provides superior physical protection and can be part of the earthing system.
- **Rules:**
 - **Conduit Fill:** You cannot fill a conduit 100% with wires. It will overheat. A maximum "fill ratio" (e.g., 40%) must be observed to allow heat to dissipate.
 - **Fittings:** All connections must be made with proper fittings (couplings, bends, connectors) and glued/secured.
- 2. **Trunking (Ducting):**
 - **Function:** A square or rectangular, lidded channel (usually PVC or metal).
 - **Use:** Common in inverter/battery rooms. It provides a clean, accessible "highway" for routing many cables neatly along a wall.
- 3. **Cable Trays:**
 - **Function:** A "ladder-like" metal tray used in large commercial/industrial installations (e.g., on a factory roof) to support hundreds of heavy cables.

4.3 WIRE SIZING (BEYOND THE BASICS)

In Term 1, we learned that small wires cause fires. Now we look at the second problem: **Voltage Drop**.

- **Ampacity (Review):** The safe current-carrying capacity of a wire. Too small = fire.
- **Voltage Drop:** This is the *energy lost* as electricity pushes through the resistance of a wire.
 - **The Problem:** All wires have a small resistance. The longer the wire, the more total resistance.

- **Ohm's Law ($V=IR$):** This resistance causes a "drop" in voltage from the panels to the inverter.
- **Example:** Your panels are on a roof, 100m away from the inverter. You use a cheap, small wire. The panels produce 400V, but the wire's resistance "consumes" 20V. The inverter only receives 380V. You have **lost 5% of your power in the wire**. It is being wasted as heat.
- **The Solution (How to reduce Voltage Drop):**
 1. **Increase Cable Size:** A thicker wire has less resistance (like a wider pipe for water). This is the "brute force" solution, but it's expensive.
 2. **Increase System Voltage (The *Smart* Solution):** This is why large systems use high-voltage string inverters (Term 1). Power (P) = Voltage (V) x Current (I).
 - To get 4,000W:
 - 48V x 83A (Low Voltage, *Very High* Current -> Needs a thick, expensive cable)
 - 400V x 10A (High Voltage, *Very Low* Current -> Can use a small, cheap cable)
 - By increasing the voltage, we dramatically lower the current, which in turn *dramatically* lowers the voltage drop.

4.4 ADVANCED CONNECTION TECHNIQUES

- **Strain Relief & Drip Loops:**
 - Wires should never be pulled "guitar-string tight" into a terminal.
 - **Drip Loop:** A small U-bend in the wire *before* it enters a junction box or building. This ensures that rainwater drips off the bottom of the loop and cannot run *into* the connection.
 - **Strain Relief:** The loop also provides "strain relief," so vibration or wind doesn't pull the wire out of its terminal.
- **Terminal Lugs and Crimping:**

- **Problem:** You cannot connect a thick battery cable (e.g., 35mm² or 50mm²) by just twisting the strands and putting them under a screw. It will be loose, create resistance, get dangerously hot, and melt.
- **Solution:** You must use a **Terminal Lug** (a copper connector). The bare wire is inserted into the lug.
- **The Tool:** A **Hydraulic Crimping Tool** is used to permanently "crimp" (compress) the lug onto the wire with several tons of pressure. This creates a perfect, cold-weld connection that is stronger than solder.
- **A bad crimp (e.g., using pliers and a hammer) is the #1 cause of battery-room fires.**
- **Junction Boxes:**
 - All outdoor wire-to-wire connections *must* be made inside a **waterproof (IP67 or higher) junction box**.
 - You can *never* use "insulation tape" and leave a connection exposed to the rain.

EVALUATION

1. What is the purpose of a conduit? Why can you not use indoor (white) PVC conduit on a sunny roof?
2. What is the difference between a conduit and trunking?
3. Explain "Voltage Drop." What are the two (2) main ways to reduce it?
4. What is a "drip loop," and what are its two functions?
5. What tool *must* be used to attach a terminal lug to a thick battery cable, and why?

ASSIGNMENT

1. Go to the main distribution board (breaker box) in your home or school.
2. Observe how the wires enter the box. Are they protected by conduit? Are they neat?
3. Take a photo (if allowed) or draw a sketch.
4. Write a one-page "Installer's Report" on the quality of the cable management. What is done well? What could be improved using the principles of conduit, trunking, and labeling?

WEEK 5: MIDTERM TEST

This week is for the midterm test, which will cover all topics from Weeks 1-4.

- **Week 1:** Solar Water Pumping (Components, TDH, Flow Rate, Sizing, Battery-free design).
- **Week 2:** Solar Street Lighting (Components, Dusk-to-Dawn, Maintenance, Failure points).
- **Week 3:** Solar Home Systems (SHS, Plug-and-Play, DC loads, PAYG model).
- **Week 4:** Advanced Wiring (Conduits, Trunking, Voltage Drop, Crimping, Drip Loops).

No new lesson notes this week.

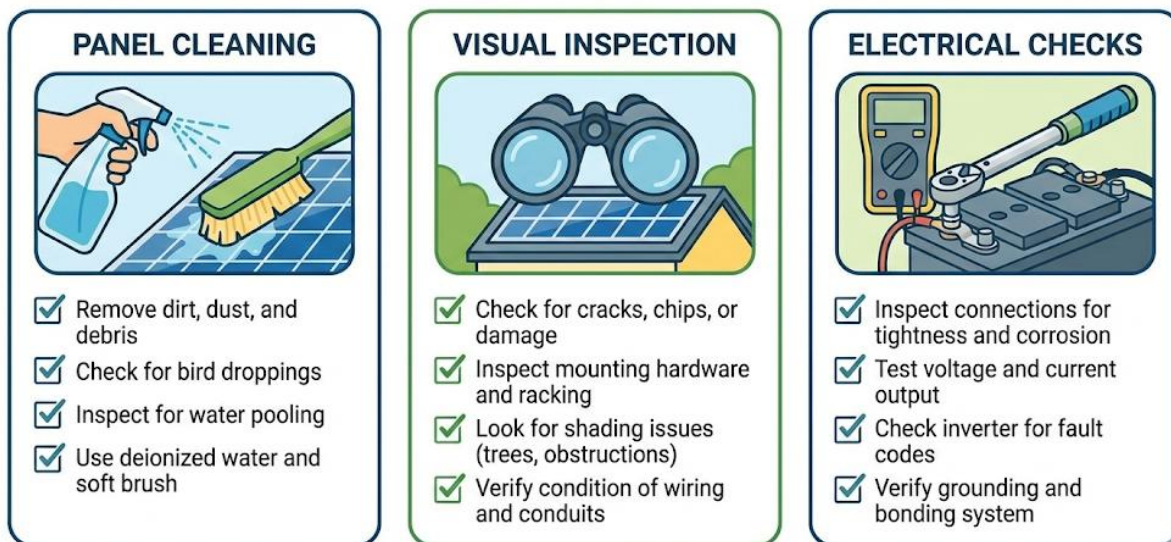


WEEK 6: PV SYSTEM MAINTENANCE I (PREVENTIVE)

6.1 INTRODUCTION

An installed PV system is not a "fit and forget" appliance. It is a power plant that, like a *Keke* or *Okada*, requires regular maintenance to perform safely, efficiently, and reliably for its 25+ year lifespan. **Preventive Maintenance (PM)** is a scheduled routine of checks and actions taken to *prevent* problems *before* they happen. It is always cheaper than **Corrective Maintenance** (fixing a broken system).

ANNUAL PV SYSTEM MAINTENANCE CHECKLIST



6.2 THE PM SCHEDULE

A professional PM visit should be scheduled at least **annually** (once per year).

- **Best Time:** At the end of the dry season / Harmattan, just before the rains. This is when systems are dirtiest and have been most stressed by heat.

6.3 KEY PREVENTIVE MAINTENANCE TASKS

1. PANEL CLEANING

- **The Problem:** Dust, grime, bird droppings, pollen, and (in Lagos) salt spray build up on the glass. This is called "**soiling**."
- **The Impact:** Soiling blocks sunlight and can reduce a panel's output by 10-40%. This *starves* the system of energy, forcing the battery to work harder and die faster.
- **Procedure:**
 - **Best Time:** Early morning or late evening. **NEVER** clean panels in the hot midday sun. The "thermal shock" of cold water on extremely hot glass can cause micro-cracks.
 - **Tools:**
 - **Correct:** Soft-bristled brush (on a long pole), clean water, a squeegee, and (if needed) a non-abrasive soap (like car wash soap).
 - **Incorrect:** **NEVER** use a hard brush, a metal scraper, or harsh detergents (like Omo or Jik). This will scratch the panel's anti-reflective coating.
 - **Method:** Wet the panels, gently scrub with the soft brush, and rinse thoroughly with clean water.

2. VISUAL INSPECTION

Walk the entire site and *look* for problems.

- **PV Array:**
 - Check for cracked, broken, or "delaminated" (peeling) panels.
 - Check for "shading." (Has a tree grown? Has the neighbour built a new wall or installed a new water tank?)
 - Check that all mounting bolts are still tight and not rusted.
 - Check that all wiring (conduits) is still secure and not cracked or broken.
- **Inverter/Battery Room:**
 - Check for signs of water leaks from the roof.
 - Check for signs of pests (rats, lizards, insects) which *love* to build nests in warm inverters. Install mesh/netting to block vents.
 - Check that inverter cooling fans are spinning and that vents are clear of dust and debris (use a vacuum or air blower).

3. ELECTRICAL CHECKS (THE MOST IMPORTANT)

- **The Problem:** Over time, heat (expansion/contraction) and vibration can cause electrical connections (terminals) to become **loose**.
- **The Danger:** A loose connection creates high **resistance**. Resistance creates **heat**. Heat melts the terminal, which causes a fire. This is the **#1 cause of electrical fires**.
- **Procedure (System MUST be SHUT DOWN - LOTO):**
 1. **LOTO (Lockout/Tagout):** The system must be *shut down* in the correct, safe sequence (AC off, then DC off).
 2. **Torque Wrench:** Use a **torque wrench** (a special screwdriver or spanner that "clicks" at a set tightness). Check *every single screw* in the system:
 - Breakers and fuses in the combiner box.
 - Battery terminals.
 - Inverter terminals (AC and DC).
 3. **Battery Terminals:**
 - Check for tightness.
 - Check for **corrosion** (a white, blue, or greenish powder).
 - **Cleaning:** Disconnect the terminal, clean it with a wire brush and a solution of baking soda and water, let it dry, and re-attach it. A light coat of battery terminal grease can prevent future corrosion.

EVALUATION

1. What is "Preventive Maintenance (PM)"?
2. What is "soiling," and how does it affect a PV system?
3. What is the *wrong* way to clean a solar panel (list 2 wrong tools/materials)?
4. Why must all electrical terminals be checked for tightness? What is the danger of a loose terminal?
5. What tool is used to properly tighten terminals to the manufacturer's specification?

ASSIGNMENT

1. Develop a one-page "Annual Maintenance Checklist" for a homeowner with a solar system.
2. The checklist should have three sections:
 - **Visual Checks** (e.g., "Check for shading," "Check for pests")
 - **Cleaning** (e.g., "Clean panels," "Clear inverter vents")
 - **Electrical Checks (To be done by a professional)** (e.g., "Torque terminals," "Check battery for corrosion")
3. List at least 4 items to check under each section.



WEEK 7: MIDTERM BREAK

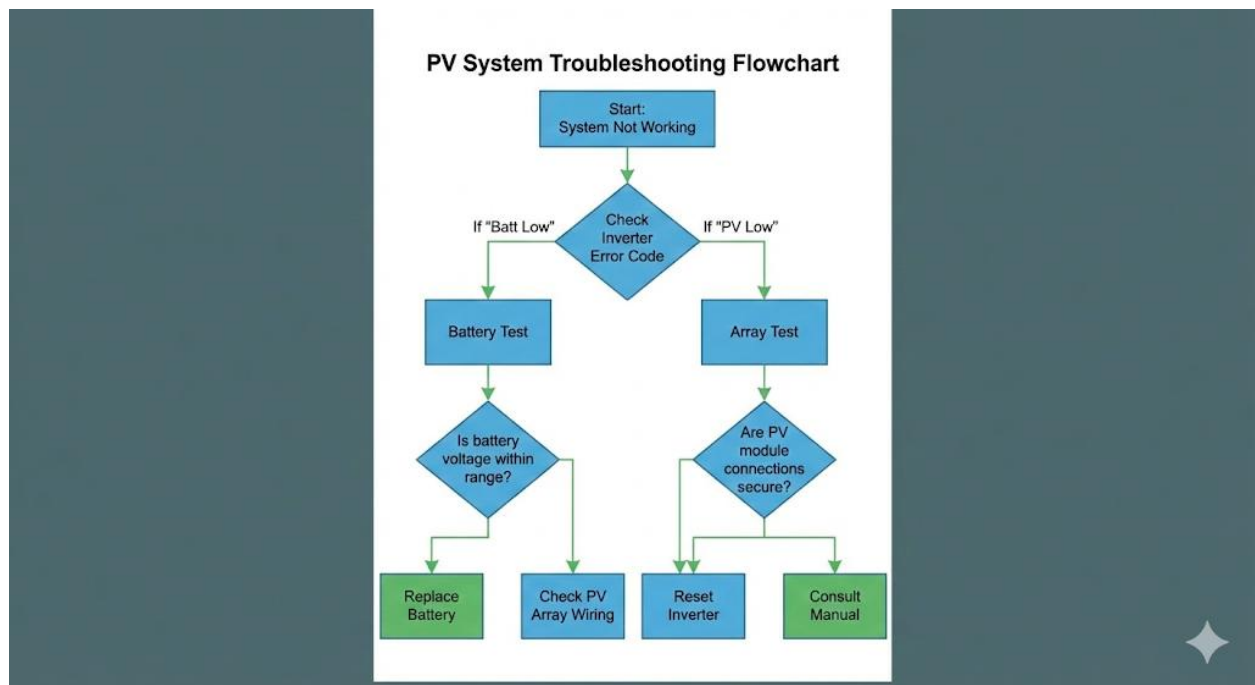
This week is the midterm break. No new lesson notes.



WEEK 8: PV SYSTEM MAINTENANCE II (TROUBLESHOOTING)

8.1 INTRODUCTION

Troubleshooting (Corrective Maintenance) is what you do when the system is *already broken*. The client has called, and they are not happy. Troubleshooting is the logical, step-by-step process of diagnosing a *fault* and repairing it.



8.2 THE LOGICAL PROCESS

Always start with the simplest, most obvious problem first. Don't go on the roof until you have checked the inverter. **Problem: "My system is not working! No power."**

STEP 1: GATHER INFORMATION & CHECK THE OBVIOUS

- **Ask the client:** "What happened? Did it go off suddenly? Has it been getting worse for weeks? Is the inverter beeping? Is there an error code on the screen?"
- **Check the Obvious (Client-Level):**
 - Has a breaker "tripped"? (Check all AC and DC breakers. Reset them *once*. If they trip again, there is a **short circuit**).

- Has the client's "load" changed? (e.g., "I just bought a new deep freezer and electric cooker." -> This is a *load* problem, not a *system* problem).
- **Check the Inverter Screen:** The inverter is the "brain." It will tell you what is wrong.
 - Error Code: "Err 05: Fan Lock" -> The fan is broken.
 - Error Code: "Err 10: Batt Low" -> The battery is dead.
 - Error Code: "Err 21: PV Low Voltage" -> No power from the panels.

STEP 2: IS IT A BATTERY PROBLEM? (The most likely culprit)

- **Symptom:** The system works perfectly fine during the day, but dies *immediately* the sun goes down (or only lasts for 1 hour).
- **Diagnosis:** The battery is dead (has reached the end of its cycle life) and is no longer holding a charge.
- **Tool:** Multimeter.
 - A healthy 12V battery should read ~12.7V (full) and ~12.0V (empty).
 - A dead 12V battery will read ~10.5V or less *and* the voltage will "crash" (drop instantly) as soon as you apply a load.
- **Solution:** Replace the battery bank.

STEP 3: IS IT A PANEL/ARRAY PROBLEM? (No DC power)

- **Symptom:** The system is off. The battery is low, *even during a sunny day*. The inverter shows a "PV Low Voltage" or "PV Input 0V" error.
- **Diagnosis:** Power is not coming from the panels.
- **Tools:** Multimeter.
- **Troubleshooting (LOTO - System OFF):**
 1. **Check for Shading/Dirt:** (See Week 6).
 2. **Check for Open Circuit:** Go to the DC disconnect or combiner box. Disconnect the main string(s).
 3. **Test for Voltage (Voc):** Set multimeter to DC Voltage.
 - Test the "Open Circuit Voltage (Voc)" of the string.
 - An array of 10 panels (20V Voc each) in series should read ~200V.

- **If $V_{oc} = 0V$:** There is a break in the string. A wire is cut, or an MC4 connector is loose. You must go to the roof and check every connection.
- 4. **Test for Short Circuit (I_{sc}):** (More advanced). Use a **Clamp Meter** to check the "Short Circuit Current (I_{sc}).\" If the current is zero, there is a problem.
- 5. **Bypass Diode Failure:** If one cell in a panel is shaded or faulty, "bypass diodes" *should* activate to let current flow "around" it. If the diode fails, it can take out 1/3 of the panel. This often shows up as a "hot spot" on a thermal camera.

STEP 4: IS IT AN INVERTER PROBLEM? (The "brain" is dead)

- **Symptom:** Panels are good (V_{oc} test is fine). Battery is good (voltage is 48V). But the inverter is "dead," no lights, no screen, no power.
- **Diagnosis:** The inverter itself has failed. This could be a blown internal fuse, a failed component (capacitor), or a major failure.
- **Solution:** Most inverters are "black boxes" and not field-serviceable. The unit must be sent back to the manufacturer or a certified repair centre.

8.3 REPLACING COMPONENTS SAFELY

- **Batteries:** When replacing batteries in a string (e.g., 4 x 12V), you must replace **all** of them. **Never mix old and new batteries.** The new batteries will be "dragged down" and destroyed by the old ones in a few months.
- **Panels:** If you replace a panel in a string, it *must* have the same (or very similar) Voltage (V_{mp}) and Current (I_{mp}) specifications as the others.

EVALUATION

1. What is the difference between Preventive and Corrective Maintenance?
2. What is the first, most obvious thing to check if a system stops working?
3. What is the most common symptom of a dead (end-of-life) battery?
4. How do you test if a solar array is producing power? What does " V_{oc} " mean?
5. Why should you never mix old and new batteries in a battery bank?

ASSIGNMENT

1. Your client calls. They say, "My system stopped working!"
2. Write a list of the *first five questions* you would ask them over the phone *before* you drive to their site.
3. (e.g., "1. Is there any error code on the inverter screen?")



WEEK 9: MONITORING & PERFORMANCE EVALUATION

9.1 INTRODUCTION

You cannot manage what you cannot measure. Monitoring is the process of using meters and software to evaluate a system's performance. This is essential for:

- Verifying the system is working as designed.
- Calculating financial savings.
- Spotting faults early (e.g., soiling) before they become major problems.



9.2 BASIC MONITORING (METERS)

1. Multimeter (The "Doctor's Stethoscope"):

- The most essential *diagnostic* tool for an installer during a maintenance visit.
- Used for *active* troubleshooting (see Week 8).
- Measures:
 - **AC/DC Voltage (V):** (Is the panel/battery/grid "on"?)
 - **AC/DC Current (A):** (How much power is flowing?)

- **Resistance (Ω):** (Is a wire broken? Is a fuse blown?)
- **Continuity (Beep Test):** (Checks if a wire is broken).

2. Clamp Meter:

- A multimeter with a "claw" or "clamp" at the top.
- **Function:** It can measure *current (Amps)* simply by clamping *around* a live wire, without needing to disconnect it. This is extremely safe, fast, and convenient.
- **Use:** Essential for checking the current from a PV string (DC) or the AC output of an inverter.

3. Inverter Display:

- The built-in screen on the inverter. This is the client's main monitoring tool.
- **It shows (in real-time):**
 - PV Voltage (V) and Current (A)
 - Battery Voltage (V) and State of Charge (SoC %)
 - AC Load (W or kW)
 - Energy produced today (kWh)
 - Error codes

9.3 ADVANCED MONITORING (SOFTWARE)

1. Data Loggers (The "Wi-Fi Dongle"):

- These are devices (e.g., a small "dongle" or built-in Wi-Fi) that connect to the inverter.
- **Function:** The logger "logs" (records) the system's performance (e.g., PV power, battery SoC, load) every few seconds and uploads it to the internet via the client's home Wi-Fi.

2. Monitoring Software (Web Portal / Mobile App):

- The manufacturer (e.g., Victron, SMA, Huawei, Growatt) provides a free app or website.
- **For the Client:** The client can log in from their phone (from anywhere in the world) and see:
 - A "flow diagram" of their house (e.g., "2kW from PV, 1kW to Load, 1kW to Battery").

- How much energy they produced today/this month.
- Their "self-consumption" rate.
- **For the Installer (The "Pro" Portal):**
 - The installer has a "pro" portal where they can see *all* their clients' systems in one dashboard.
 - **Benefit:** This is proactive, professional maintenance. You can spot a problem *before* the client even knows. You can see a "PV Low Voltage" alert and call the client, "Good morning, Mrs. B, I see your panels are under-performing. They are probably dirty. Let's schedule a cleaning."

9.4 PERFORMANCE EVALUATION

- **Performance Ratio (PR):**
 - This is the key metric for large-scale commercial systems.
 - **Definition:** A measure of the *quality* and *efficiency* of an installation, from 0% to 100%.
 - **Formula (Simplified):** $PR = (\text{Actual Energy Output}) / (\text{Theoretical Max Output})$
 - **What it means:** A PR of 85% means that 85% of the sunlight energy (after panel efficiency) was successfully converted and delivered as AC electricity. The other 15% was lost to "system losses" (heat, wire resistance, inverter inefficiency, dust, etc.).
 - A "good" PR is 80-90%. A "bad" PR (e.g., 60%) means something is wrong (e.g., massive soiling, bad inverter, major shading).

EVALUATION

1. What is the main advantage of a Clamp Meter over a standard Multimeter for measuring current?
2. What is a "Data Logger," and what does it use to connect to the internet?
3. List three (3) benefits of a software monitoring portal for the *installer*.
4. What is the "Performance Ratio (PR)"?
5. What would a very low PR (e.g., 55%) indicate?

ASSIGNMENT

1. Download a solar monitoring app on your phone (e.g., "Victron VRM," "Huawei FusionSolar"). Many of them have a "Demo" login.
2. Explore the demo system. Find the "flow diagram" and the "historical data" (e.g., energy produced yesterday).
3. Write a short paragraph describing what you learned from the app.



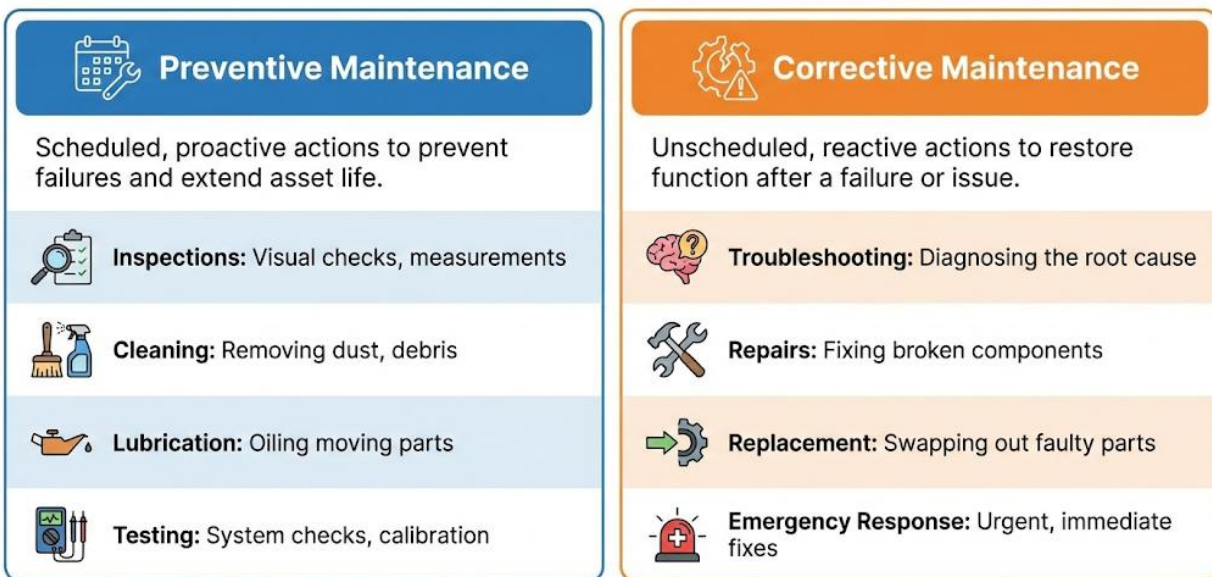
WEEK 10: PRACTICAL: MAINTENANCE OF INSTALLED SYSTEMS

10.1 OBJECTIVE

To apply the knowledge from Weeks 6, 8, and 9 in a hands-on, practical maintenance and troubleshooting session on an installed (or simulated) PV system.

!! SAFETY WARNING !! All work on live systems must be supervised by a qualified instructor. All students must wear appropriate PPE (safety glasses, insulated gloves). LOTO (Lockout/Tagout) procedures *must* be followed before any electrical work.

Preventive vs. Corrective Maintenance



10.2 PRACTICAL 1: PREVENTIVE MAINTENANCE (PM)

- **Objective:** To conduct a full annual PM routine.
- **Tools:** PM checklist (from Week 6 assignment), cleaning kit, multimeter, torque wrench set.
- **Procedure:**

1. **Panel Cleaning:** (Simulated or actual). Students identify the correct tools (soft brush, water) and discuss the best time (early morning) and method (no harsh soap).
2. **Visual Inspection:** Students inspect the array and inverter room, checking for:
 - Loose bolts on the mounting rack.
 - Signs of shading.
 - Pest/water ingress in the inverter room.
 - Cable management (are all conduits and trunking secure?).
3. **Electrical Checks (Supervised):**
 - Instructor demonstrates the LOTO shutdown sequence (AC loads -> AC breakers -> DC disconnect).
 - Instructor demonstrates checking the torque on a battery terminal and an AC breaker using a torque wrench.
 - Students (wearing gloves) practice cleaning a (disconnected) corroded battery terminal with a wire brush and baking soda solution.

10.3 PRACTICAL 2: TROUBLESHOOTING A FAULT (FAULT-FINDING)

- **Objective:** To diagnose a fault planted by the instructor.
- **Fault Scenario 1: "Dead Battery"**
 - **Instructor:** (Prepares the system with a known-bad battery). "The client says the system works in the day but dies at night."
 - **Students:** (Under supervision)
 1. **Hypothesis:** The battery is the likely fault.
 2. **Test:** Use a multimeter to check the battery's "resting voltage" (e.g., it reads 11V).
 3. **Confirm:** Apply a load (turn on the inverter) and watch the voltage "crash" (e.g., to 9V).
 4. **Diagnosis:** Battery is at end-of-life. **Solution:** Replace battery bank (all units).
- **Fault Scenario 2: "Open Circuit Array"**

- **Instructor:** (Prepares the system by disconnecting one MC4 connector in the array string). "The client says it's a sunny day, but the battery is not charging."
- **Students:** (Under supervision)
 1. **Hypothesis:** No power from the panels.
 2. **Test 1 (Inverter):** Check the inverter display (it shows "PV Low Voltage" or 0V).
 3. **Test 2 (Multimeter):** (LOTO procedure). Go to the DC disconnect. Use a multimeter to test the "Open Circuit Voltage (Voc)" from the array.
 4. **Observation:** The multimeter reads 0V (or very low).
 5. **Diagnosis:** There is a break in the string. **Solution:** Go to the roof and check every panel and connector until the open circuit is found.
- **Fault Scenario 3: "Tripped Breaker"**
 - **Instructor:** (Trips the main AC breaker for the inverter). "The client says the system suddenly went dead."
 - **Students:**
 1. **Hypothesis:** Simplest thing first.
 2. **Test:** Check the distribution board.
 3. **Observation:** The breaker is tripped.
 4. **Diagnosis:** Overload or short circuit. **Solution:** Ask the client what they just plugged in (e.g., a new water heater). Reset the breaker.

EVALUATION

1. What is the first step before performing any electrical checks on a PV system?
2. What tool is used to check the tightness of a battery terminal to the correct specification?
3. What is the "Voc" of a solar array?
4. If you test the Voc of an array and it reads 0V, what is the likely problem?
5. What is the main symptom of a "dead" (end-of-life) battery?

ASSIGNMENT

1. Write a formal "Maintenance Report" for *one* of the fault scenarios from today's practical.

2. The report should be one page and include:

- Client's Complaint (e.g., "System dies at night")
- Diagnostic Steps Taken (e.g., "Checked battery voltage at rest...")
- Fault Found (e.g., "Battery bank at end-of-life")
- Recommended Solution (e.g., "Replace all 4 batteries")
- Parts/Labour Required (e.g., "4x 12V 200Ah batteries, 2 hours labour")



WEEKS 11–13: REVISION & EXAMS

11.1 SUMMARY OF TERM 2

This term, we focused on the practical *applications* and *long-term maintenance* of PV systems.

- **Applications:** We learned to design and size three specific, high-impact systems:
 - **Solar Water Pumps:** (TDH, Flow Rate, battery-free design for agriculture).
 - **Solar Street Lights:** (Dusk-to-dawn, maintenance, common failures).
 - **Solar Home Systems (SHS):** (Plug-and-play, PAYG models for energy access).
- **Installation Quality:** We learned advanced wiring techniques, including **conduits/trunking**, **voltage drop** calculations, and **crimping**.
- **Maintenance (Preventive):** We learned the routine PM schedule, focusing on **panel cleaning** and **terminal checking (torque)**.
- **Maintenance (Corrective):** We learned the logical troubleshooting process for identifying faults in the **battery**, **PV array**, or **inverter**.
- **Monitoring:** We learned the difference between basic (multimeter) and advanced (software) monitoring and how to evaluate **Performance Ratio (PR)**.

11.2 FINAL EXAM PREPARATION

- Be ready to size a simple solar water pump (calculate array size from pump power).
- Be able to list the components and maintenance needs of a solar street light.
- Be able to explain the PAYG model.
- Be able to list the 4-step process for troubleshooting a "dead" system.
- Be able to explain the difference between Preventive and Corrective maintenance.

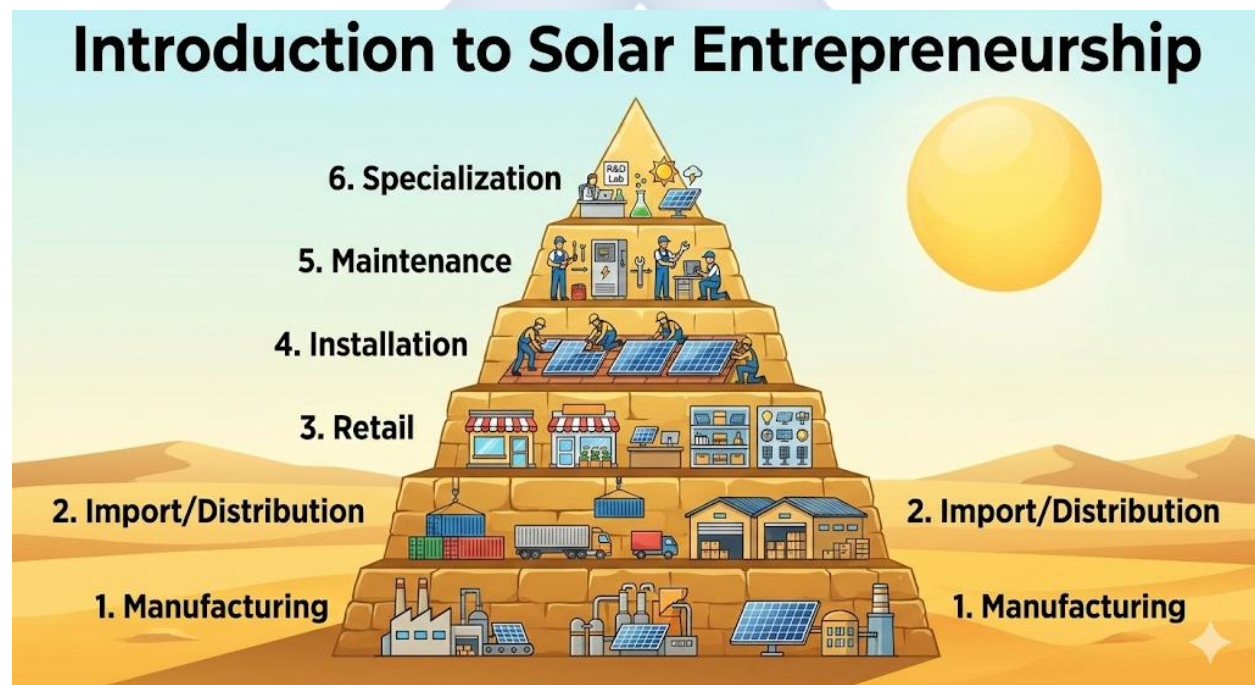
SS2 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

THIRD TERM

WEEK 1: INTRODUCTION TO SOLAR ENTREPRENEURSHIP

1.1 INTRODUCTION

Welcome to the Third Term. For the past two terms, we have focused on the *technical* skills of solar installation: how to design a system, how to size it, and how to maintain it. This term, we transition to the *business* of solar. A great installer who doesn't understand business will fail, while a good installer with excellent business skills can build an empire.



Entrepreneurship is the process of designing, launching, and running a new business. It involves identifying a market opportunity (a problem) and creating a solution (a product or service) to meet that need, often in the face of risk and uncertainty, with the goal of making a profit.

1.2 THE NIGERIAN OPPORTUNITY: THE PROBLEM & THE SOLUTION

The opportunity for solar entrepreneurship in Nigeria is one of the largest in the world, driven by a simple, massive gap between energy *demand* and energy *supply*.

THE PROBLEM (THE "WHY"):

1. **The Grid Deficit:** The national grid (NEPA/PHCN) is unreliable, with an estimated total capacity of ~12,500 MW but often delivering only ~4,000 MW to a population of 200+ million people. This is one of the lowest per-capita power consumptions in the world.
2. **The Cost of "Alternatives":** The *real* power grid in Nigeria is the sound of generators. Businesses and homes spend billions of naira (trillions, collectively) annually on diesel and petrol generators. This is:
 - **Expensive:** Diesel/petrol prices are high and volatile.
 - **Noisy:** Causes significant noise pollution.
 - **Dirty:** Causes air pollution and health problems.
 - **High-Maintenance:** Generators require constant servicing, oil changes, and parts.
3. **The "Energy Ladder":** At the lowest-income level, families spend a high *percentage* of their income on "dirty" energy like kerosene, candles, and charging phones at a neighbour's shop.

THE SOLUTION (THE "OPPORTUNITY"):

Solar power directly solves these problems.

1. **Abundant Resource:** Nigeria is geographically blessed with "Peak Sun Hours" (PSH) averaging 4.5 to 6.5 hours per day across the country. The "fuel" is free, abundant, and reliable.
2. **Falling Costs:** The global price of solar panels (in \$/watt) has dropped by over 90% in the last decade.
3. **Rising Cost of Alternatives:** As grid tariffs and fuel costs increase, the "payback period" for a solar system becomes shorter and shorter.
4. **Scalability:** The opportunity is not just "one size fits all." It exists at every level, from a 20W "Solar Home System" to a 500MW utility-scale solar farm.

1.3 THE SOLAR VALUE CHAIN: WHERE CAN YOU FIT IN?

"Solar entrepreneurship" does not just mean "installer." There are many business models within the value chain:

1. Tier 1: Manufacturing / Assembly:

- This is the most capital-intensive. It involves setting up factories to assemble solar panels, inverters, or batteries locally (e.g., the few panel assembly plants in Nigeria).

2. Tier 2: Importation & Distribution (Wholesale):

- These are the large-scale "importers" who bring in containers of equipment (e.g., from Victron, SMA, Growatt, JA Solar) and sell them in bulk to smaller companies. This requires significant capital and logistics expertise.

3. Tier 3: Retail & E-Commerce:

- These are the "shops" (physical or online) that sell components directly to the public and to small installers. They hold inventory.

4. Tier 4: Solar Installation (System Integration):

- This is the *core* of the industry and our focus. The installer is the "solution provider." They design and build systems for a client. This is a *service* business, not just a product business. It requires the least startup capital but the most *technical skill*.

5. Tier 5: Maintenance & Service:

- This is a highly profitable and often overlooked niche. It involves offering "after-sales" service, such as annual maintenance contracts (AMCs), panel cleaning, and battery replacement for *existing* systems (even those installed by other, less professional companies).

6. Tier 6: Specialization:

- Becoming an expert in *one specific area*, such as:
 - Solar Water Pumping (for agriculture).
 - Mini-Grids (for rural electrification).
 - Commercial & Industrial (C&I) installations (for banks and factories).
 - PAYG (Pay-As-You-Go) Solar Home Systems.

1.4 CHALLENGES AND RISKS

Entrepreneurship is not easy. A successful solar entrepreneur must be prepared for:

- **High Competition:** It's a growing market, and many people are rushing in. You must compete on *quality*, not just price.
- **Sub-standard Products:** The market is flooded with "fake" or low-quality panels, batteries, and inverters. A key skill is **procurement**—knowing how to source genuine products.
- **Startup Capital:** While less than manufacturing, you still need money for tools, transportation, and potentially a small office.
- **Customer Education:** Many clients have unrealistic expectations (e.g., wanting to power 3 ACs with a 1kVA system) and need to be educated.

EVALUATION

1. Define "Entrepreneurship" in the context of the solar industry.
2. List three (3) major problems in the Nigerian power sector that create an opportunity for solar.
3. What is the "solar value chain"? List four (4) different business models within it.
4. What is the difference between an "Installer" (Tier 4) and a "Distributor" (Tier 2)?
5. What are two (2) major challenges a new solar entrepreneur will face in Nigeria?

ASSIGNMENT

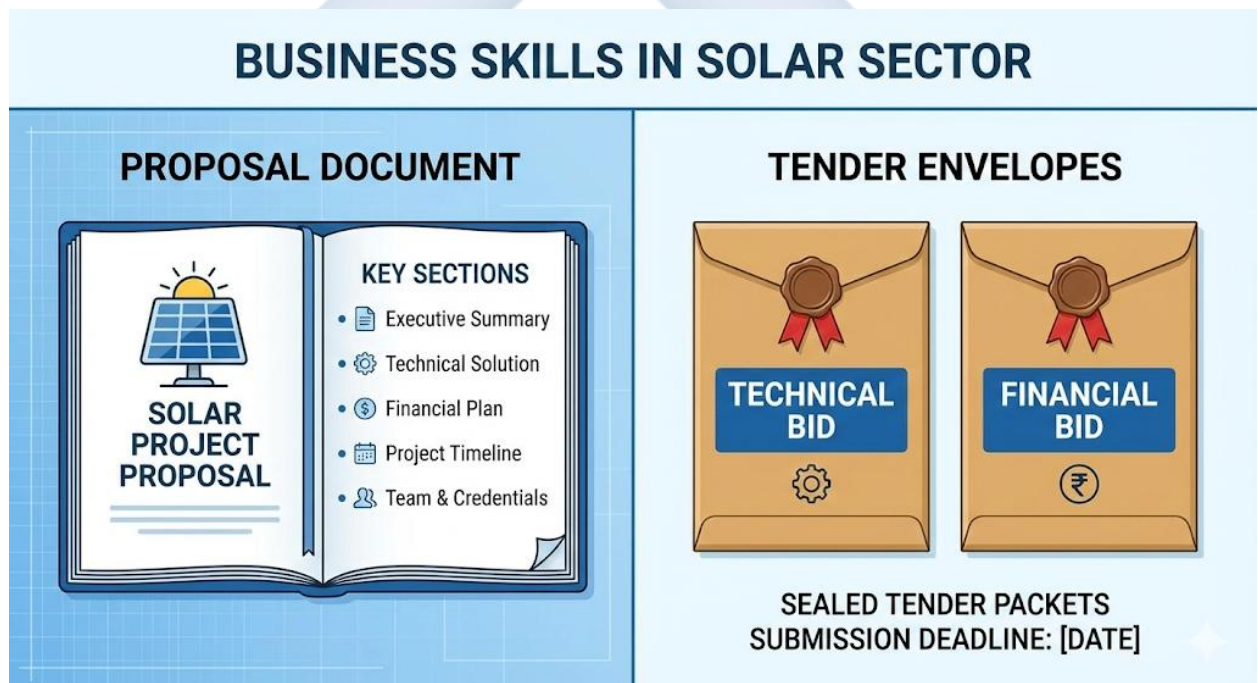
1. Interview a local solar installer or a shop owner who sells solar components (even a small one).
2. Ask them three questions: a) What is the biggest opportunity they see in the market? b) What is their single biggest challenge (e.g., bad batteries, finding clients, government policies)? c) What one piece of advice would they give to someone starting out?
3. Write a one-page report on your findings.

WEEK 2: BUSINESS SKILLS IN SOLAR SECTOR

2.1 INTRODUCTION

Having a technical skill (like solar installation) is different from having a *business*. To be an entrepreneur, you must learn to *get* the job. The two most important skills for getting jobs are proposal writing and tendering.

- **Proposal:** A sales document, usually written for a *private* client (homeowner, small business) who you have spoken with.
- **Tender (Bid):** A formal, legal document, usually submitted in a sealed envelope in response to a *public* advertisement (e.g., from the government, a university, or a large corporation).



2.2 PROPOSAL WRITING

A proposal is your chance to show the client you understand their problem and are the best person to solve it. It must be professional, clear, and convincing.

THE STRUCTURE OF A PROFESSIONAL PROPOSAL:

1. **Cover Letter:** A one-page introduction on your company letterhead.
 - "Dear Mr. Adebayo, Thank you for the opportunity to provide a solar solution for your home in Ikeja. We are confident we have the perfect system for your needs..."
2. **Executive Summary:** A single paragraph that summarizes the *entire* proposal. (The problem, our solution, the cost).
3. **Understanding the Need (The Load Audit):**
 - This is where you *prove* you listened.
 - "Based on our site survey, we understand your goal is to power the following loads..."
 - List the client's loads (lights, fans, TV, fridge) and their estimated daily energy need (e.g., 10 kWh/day).
4. **The Proposed Solution (The Technical Specification):**
 - This is the *technical* core.
 - **Inverter:** "We propose a 5kVA/48V Hybrid Inverter (Brand: Victron)..."
 - **Battery Bank:** "We propose a 10.2kWh Lithium-ion (LiFePO4) Battery..."
 - **PV Array:** "We propose 6kWp of PV panels (12 x 550W Monocrystalline panels)..."
 - **Mounting:** "The array will be installed on a high-quality aluminum roof-mount racking system."
5. **System Performance & Expectations:**
 - Manage expectations!
 - "This system is designed to provide 24/7 power for your essential loads. During the day, the system will be powered by the sun. At night, it will run from the batteries. *The system is designed to provide 6 hours of backup for your 1.5HP Air Conditioner, but heavy use will deplete the battery faster.*"
6. **Bill of Quantities (BOQ) / Financial Investment:**
 - The *price*. A detailed, itemized list of all costs (see Week 3).
 - (e.g., Inverter... ₦X; Panels... ₦Y; Batteries... ₦Z; Installation... ₦A).
 - **Total:** ₦X+Y+Z+A
7. **Project Timeline:**

- "This project will be completed within **three (3) days** of receiving the initial deposit."

8. **Warranty & After-Sales Service:**

- "We provide a 1-year warranty on our installation workmanship, 5 years on the inverter, and 10 years on the panels..."

9. **About Us / Past Projects:**

- Why should they trust you? Include your company registration number, photos of past jobs, and client testimonials.

2.3 TENDERING (BIDDING)

Tendering is a formal, high-stakes process. Large organizations (like banks or government agencies) *must* use it to ensure transparency.

- **RFP/ITB:** You will see an "Invitation to Bid" (ITB) or "Request for Proposal" (RFP) in a newspaper or online.
- **The Process:**
 1. **Find the Bid:** Actively look for them.
 2. **Purchase the Bid Document:** You often have to pay a non-refundable fee (e.g., ₦20,000) just to get the *details* of the job.
 3. **Read VERY Carefully:** The document will be 100+ pages long, full of legal and technical requirements.
 4. **Prepare the Bid Package:** This is *not* one document, but two large, separate, sealed envelopes.
 - **Technical Bid:** Your company registration (CAC), tax clearance (FIRS), PenCom certificate, past project history, technical solution, *without any prices*.
 - **Financial Bid:** The price (BOQ) *only*.
 5. **Submission:** You must submit both envelopes physically before the exact deadline (e.g., 12:00 PM, Tuesday). If you are one minute late, you are disqualified.
- **Why Tendering is Hard:**

- **It's expensive:** You can spend ₦100,000 preparing a bid you might not win.
- **It's all-or-nothing:** You are often judged on the *lowest price* (for the "Financial Bid") *after* you have "passed" the technical stage.
- **Compliance is key:** If you forget to attach your tax clearance, your entire bid is "non-responsive" and thrown in the trash.

EVALUATION

1. What is the main difference between a "Proposal" and a "Tender"?
2. List five (5) essential sections of a professional solar proposal.
3. What is the purpose of the "Understanding the Need" section?
4. Why is "Managing Expectations" important?
5. What are the "Technical Bid" and "Financial Bid" in a formal tender? Why are they separate?

ASSIGNMENT

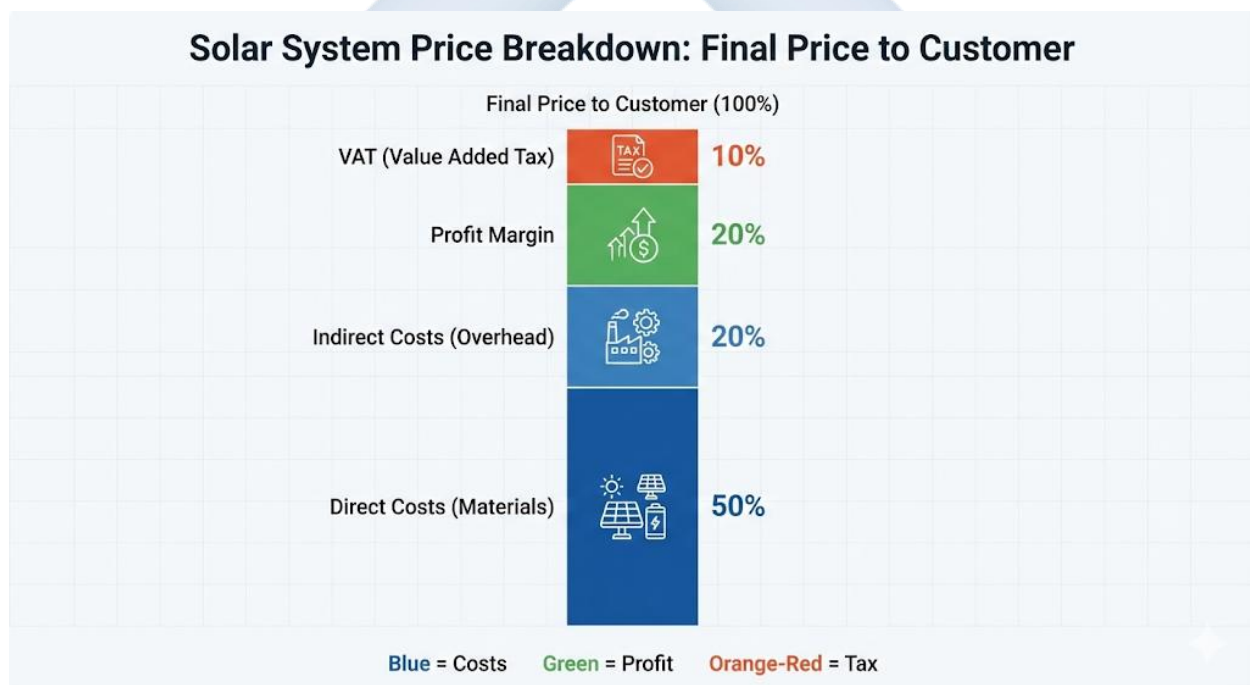
1. Find a solar proposal online (search for "Sample Solar Proposal PDF").
2. Analyze its structure. How is it similar to the one in our notes? How is it different?
3. Write a half-page critique. What did they do well? What was confusing or unprofessional?

WEEK 3: COSTING & PRICING OF SOLAR SYSTEMS

3.1 INTRODUCTION

This is the most important lesson for any business. More businesses fail from *bad pricing* than from bad technical work. You *must* know your numbers.

- **Costing:** The process of figuring out *exactly* how much a job *costs you* to deliver.
- **Pricing:** The process of deciding what to *charge the customer*, which *must* be higher than your cost.
- **Price:** What the customer pays.
- **Cost:** What you pay.
- **Profit (Margin):** The difference between Price and Cost. This is the goal.



3.2 THE BILL OF QUANTITIES (BOQ)

The BOQ (or "Bill of Materials") is the foundation of all costing. It is a detailed, itemized list of *every single component* required for the installation.

Sample BOQ for a 5kVA System:

S/N	Item	Description	Qty	Unit Cost (₦)	Total Cost (₦)
A	Main Components (Majors)				
1	Inverter	5kVA/48V Hybrid (e.g., Growatt)	1	450,000	450,000
2	Battery	5.1kWh LiFePO4 / 48V	1	800,000	800,000
3	Panels	550W Monocrystalline	8	85,000	680,000
B	Balance of System (BOS)				
4	Mounting System	Aluminum Roof Racks (full set)	1	120,000	120,000
5	DC Cable	6mm ² PV Cable (100m roll)	1	75,000	75,000
6	Battery Cable	35mm ² Cable (Red/Black)	4m	8,000/m	32,000
7	AC Cable	4mm ² (In) / 2.5mm ² (Out)	20m	1,500/m	30,000
8	Combiner Box	4-in-1-out, with Fuses & SPD	1	45,000	45,000
9	Breakers	1 x 125A DC, 1 x 32A AC	1 set	25,000	25,000
10	Conduits	25mm PVC (UV-rated)	10	1,000	10,000
11	Earthing	Rod, Cable, Clamps	1 set	18,000	18,000
12	Consumables	Lugs, Trunking, Screws	1 lot	15,000	15,000
C	Logistics & Labor				
13	Labor	1 Lead Installer, 2 Assistants (3 Days)	1 lot	90,000	90,000
14	Transport	Van hire for materials & team	1 lot	35,000	35,000

3.3 COST ANALYSIS: THE FULL CALCULATION

The BOQ is *not* your total cost.

Step 1: Find your Direct Costs (Cost of Goods Sold - COGS)

- This is the total of your BOQ.
- **Direct Cost = ₦2,425,000**

Step 2: Add your Indirect Costs (Overhead)

- This is the "hidden" cost of *running your business*, which must be shared among all your jobs.
- *Your Monthly Overhead:* Office Rent (₦50k), your *own* Salary (₦150k), Internet/Phone (₦20k), Marketing (₦30k) = **₦250,000 / month.**
- *Job Prorating:* If you do 5 jobs this month, you must add $(₦250,000 / 5) = \text{₦50,000}$ to *each job.*
- **Total Cost = Direct Cost + Overhead**
- **Total Cost = ₦2,425,000 + ₦50,000 = ₦2,475,000**
- (*This is your "break-even" price. If you charge this, you make ₦0 profit.*)

Step 3: Add your Profit Margin

- This is the *company's* earning, which allows it to grow.
- A healthy margin is 15% - 30%. Let's use **20%**.
- **Profit = Total Cost x 20%**
- **Profit = ₦2,475,000 x 0.20 = ₦495,000**

Step 4: Calculate the Final Price (excl. VAT)

- **Final Price = Total Cost + Profit**
- **Final Price = ₦2,475,000 + ₦495,000 = ₦2,970,000**

Step 5: Add VAT (Value Added Tax)

- VAT in Nigeria is 7.5%. This is a *tax* you collect *for the government.*

- $\text{VAT} = \text{Final Price} \times 7.5\%$
- $\text{VAT} = \text{₱}2,970,000 \times 0.075 = \text{₱}222,750$
- $\text{FINAL PRICE TO CUSTOMER} = \text{₱}2,970,000 + \text{₱}222,750 = \text{₱}3,192,750$

The "Amateur" Mistake:

- Amateur adds up materials (₱1.9M), adds ₱150k for "labor and profit," and charges **₱2,050,000**.
- They *lost* ₱425,000 on the job and don't even know it. This is why they go out of business.

EVALUATION

1. What is the difference between "Costing" and "Pricing"?
2. What is a "Bill of Quantities" (BOQ)?
3. What are "Direct Costs" vs. "Indirect Costs (Overhead)"? Give one example of each.
4. Why is it a mistake to *only* add up material costs when setting a price?
5. What is a "Profit Margin"?

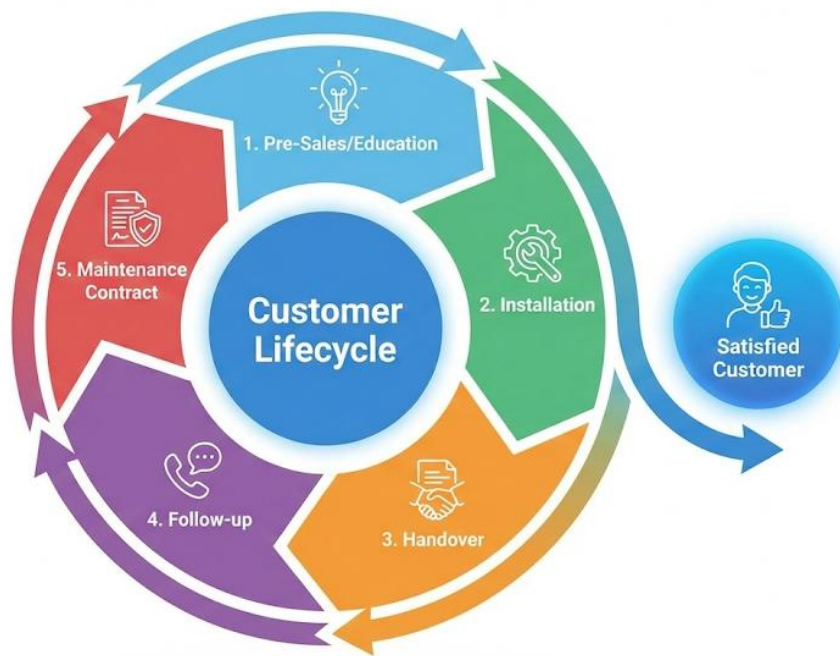
ASSIGNMENT

1. Using the sample BOQ, re-calculate the **Final Price to Customer** if you wanted to make a **30% Profit Margin** instead of 20%.
2. Show your steps: (Total Cost = ₱2,475,000), (New Profit = ?), (New Final Price = ?), (New VAT = ?), (New Final Price to Customer = ?).

WEEK 4: CUSTOMER SERVICE & SUPPORT

4.1 INTRODUCTION

Getting one job is a transaction. Getting *referrals* is a business. In the solar industry, a good reputation is your most valuable asset. Customer service is the entire experience a client has with your company—from their first phone call to the maintenance visit five years later.



4.2 THE CUSTOMER LIFECYCLE

PHASE 1: PRE-SALES & EDUCATION

- **Listen:** The client often doesn't know what they need. They might ask for a "2kVA system" but have a 2HP water pump. Your first job is to *listen* to their *problem* (e.g., "I am tired of the generator noise," "My NEPA bill is too high").
- **Energy Audit:** This is the *start* of good service. Visit their site. See the appliances.
- **Educate:** Explain the realities.
 - "No, solar cannot power your cooking."
 - "This is why lithium batteries are more expensive but *cheaper* in the long run."

- **Manage Expectations:** This is the most important skill.
 - **GOOD:** "This 5kVA system will power all your lights, fans, TVs, and fridge 24/7. It will *not* power your 3 ACs at night."
 - **BAD:** "Yes, this system will power everything!" (This client will call you, angrily, at 2 AM when their battery dies).

PHASE 2: DURING INSTALLATION

- **Professionalism:** Your team is in the client's *home*.
 - Wear uniforms or branded shirts.
 - Be polite.
 - Do *not* make a mess. Use drop-cloths, clean up all wire scraps and dust.
 - Communicate clearly: "Good morning, we are starting on the roof now. We will need to turn off the power for 1 hour at 2 PM."
- **Quality:** The client may not know a good crimp, but they know "neatness." Neat trunking and cable management *shows* them the quality they paid for.

PHASE 3: AFTER-SALES SERVICE & HANDOVER This is where you build a customer for life.

1. The Handover (Client Training):

- Do *not* just "turn it on and run." Spend 30 minutes with the client.
- **Show them the Inverter:** "This is the screen. This number is your battery percentage. This light means you are on solar..."
- **Show them the Breakers:** "This is your main DC breaker. This is your AC breaker. If there is a problem, this is what you should check first."
- **Give them the "Keys":** Provide a *User Manual* (even a simple 2-page document) with your phone number and warranty details.

2. The Warranty:

- Be *clear* about your warranty.
- "1-Year Workmanship" (This covers *your* installation - e.g., a wire comes loose).
- "Manufacturer's Warranty" (This covers the *product* - e.g., the inverter fails).

3. The Follow-Up:

- Call the client one week later. "Good morning, Mrs. Obi. This is [Your Name] from [Your Company]. I'm just calling to check in. How are you enjoying your new solar system?"
- This 2-minute call is *powerful*. It shows you care and builds immense trust.

4. Recurring Revenue (Maintenance Contracts):

- After the follow-up, offer them a service.
- "To keep your system running at 100%, we offer an Annual Maintenance Contract. For ₦X per year, we will visit twice to clean your panels and check all your connections."
- This creates a stable, predictable income stream for your business.

EVALUATION

1. Why is "managing expectations" the most important part of the pre-sales process?
2. List three (3) unprofessional behaviors an installation team might do at a client's site.
3. What is "The Handover"? What three things must you show the client?
4. What is the difference between a "Workmanship Warranty" and a "Manufacturer's Warranty"?
5. What is a "Maintenance Contract," and why is it a good business model?

ASSIGNMENT

1. Write a simple, one-page "Client Handover Sheet" / "Quick Start Guide" for a 5kVA system.
2. It should include:
 - Your (fictional) company name and emergency phone number.
 - A simple labeled diagram of the inverter screen (what the numbers mean).
 - A "What to do if..." section (e.g., "...the power goes out" -> "1. Check your breakers... 2. Call us.").

WEEK 5: MIDTERM TEST

This week is for the midterm test, which will cover all topics from Weeks 1-4.

- **Week 1:** The Nigerian solar opportunity, the value chain, and business models.
- **Week 2:** Proposal writing structure, Tendering (RFP/ITB), Technical vs. Financial bids.
- **Week 3:** Costing vs. Pricing, Bill of Quantities (BOQ), Direct vs. Indirect Costs (Overhead), Profit Margin calculation.
- **Week 4:** Customer service, managing expectations, the handover process, after-sales service, and maintenance contracts.

No new lesson notes this week.



WEEK 6: PROJECT MANAGEMENT IN SOLAR INSTALLATIONS

6.1 INTRODUCTION

If a "project" is a job with a start, a middle, and an end (the installation), then "project management" is the skill of *guiding* that job to a successful completion. A good project manager ensures the job is done **on time**, **on budget**, and **to the required quality**.

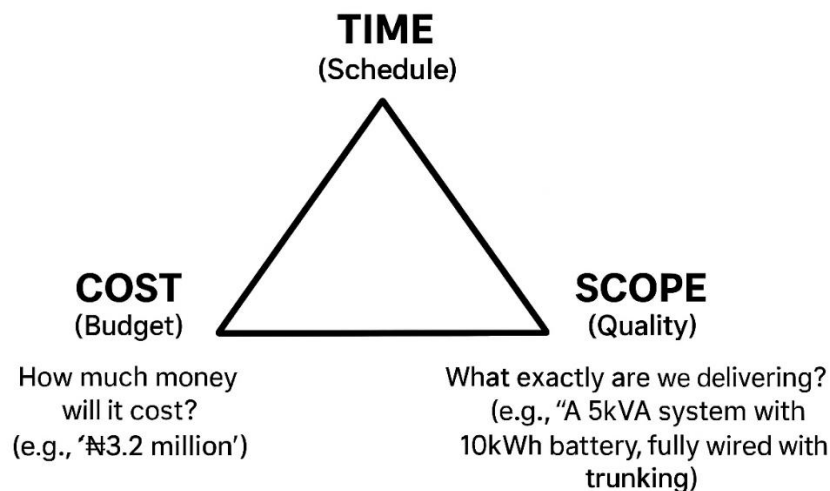
6.2 THE "IRON TRIANGLE" (THE THREE CONSTRAINTS)

This is the fundamental law of project management. Every project has three constraints that are locked together.



THE "IRON TRIANGLE" (THE THREE CONSTRAINTS)

This is the fundamental law of project management.
Every project has three constraints that are locked together.



1. **TIME (Schedule):** How long will the project take? (e.g., "3 days").
2. **COST (Budget):** How much money will it cost? (e.g., "₦3.2 million").
3. **SCOPE (Quality):** What *exactly* are we delivering? (e.g., "A 5kVA system with 10kWh battery, fully wired with trunking").

The Law: You can *never* change one point of the triangle without affecting the others.

- **Client:** "I need it done *faster!*" (Decrease Time)
- **You:** "Okay, but I'll need to hire two extra installers. That will increase the *Cost.*"
- **Client:** "I want a *cheaper* price!" (Decrease Cost)
- **You:** "Okay, but to meet that budget, we will have to reduce the *Scope* (e.g., use a smaller battery or cheaper panels)."
- **Client:** "I want to add my 2 water heaters to the system!" (Increase Scope)
- **You:** "Okay, but that will *double* the Cost (for more batteries) and *increase* the Time."

A good project manager's job is to *balance* these three points.

6.3 THE 5 PHASES OF A SOLAR PROJECT

1. PHASE 1: INITIATION

- This is the "go" signal.
- The client accepts your proposal (Price, Scope) and signs the contract.
- The client pays the initial deposit (e.g., 70%).
- The project is now *live*.

2. PHASE 2: PLANNING

- This is the *most important* phase.
- **Create the BOQ** (Week 3).
- **Procurement:** *Order and buy* all the materials (panels, inverter, etc.).
- **Resource Planning:** Assign a team (e.g., "Team B: Tunde (Lead), Femi, John").
- **Schedule:** Set the installation date (e.g., "Next Tuesday, 8 AM").
- **Logistics:** Book the transport van.

3. PHASE 3: EXECUTION

- This is the *work*.
- The team arrives on-site.
- **Day 1:** Roof-mounting, running conduits, panel installation.
- **Day 2:** Battery/Inverter mounting, DC wiring, AC wiring, Earthing.
- **Day 3:** Commissioning, testing, and client handover.

4. PHASE 4: MONITORING & CONTROL (QUALITY CONTROL)

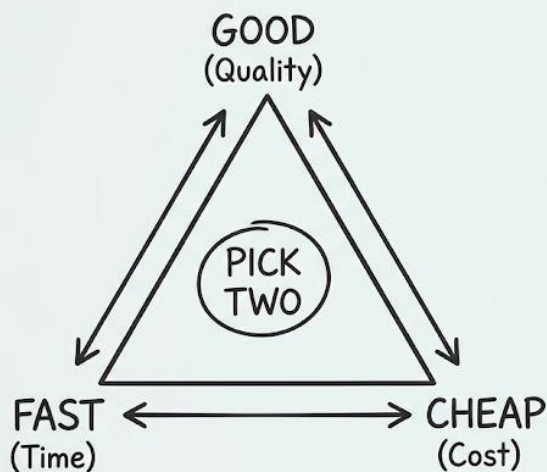
- This happens *during* Execution.

- The Team Lead (Project Manager) is *not* just working; they are *checking*.
- "Femi, is that terminal torqued?"
- "John, that trunking is not straight. Redo it."
- "Are we on schedule? It's 2 PM, we should be finished with the roof."
- "The client wants to add a new socket. (Scope change!) Stop. Call the office. We must price that separately."

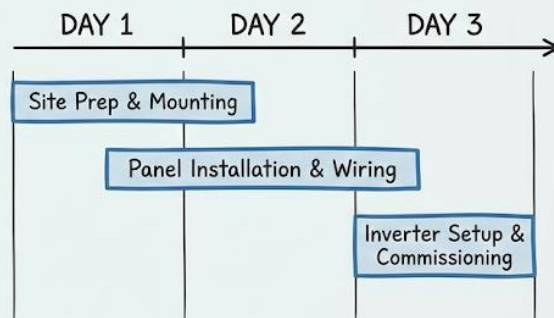
5. PHASE 5: CLOSING

- The job is done.
- The client signs a "Project Completion" form.
- You collect the final payment (e.g., 30%).
- You schedule the one-week follow-up call.
- The project is now *closed*.

PROJECT MANAGEMENT IN SOLAR INSTALLATIONS



3-DAY INSTALLATION TIMELINE



6.4 RESOURCE MANAGEMENT

- **Human Resources:** Who is on your team? A good team is *not* "3 installers." It's:
 - 1 x **Lead Installer (Technician):** The "brain." Reads the design. Manages the client.

- 2 x **Installers (Assistants):** Know how to work, run cables, mount panels.
- 1 x **Laborer (Optional):** For carrying materials, support.
- **Tool Management:** You must have a "Job Box" with the right tools. A team arriving without a hydraulic crimper or a torque wrench *cannot* do a quality job.

EVALUATION

1. What is Project Management?
2. What are the three (3) constraints of the "Iron Triangle"?
3. Give an example of how changing the "Scope" affects "Cost" and "Time."
4. What are the 5 phases of a solar project?
5. What is "Quality Control," and in which phase does it happen?

ASSIGNMENT

1. Think about the 3-day installation (Phase 3: Execution) described above.
2. Create a more detailed **Project Schedule** for the "Lead Installer" to follow.
3. *Example:*
 - *Day 1 (8 AM):* Arrive on-site, greet client, confirm inverter location.
 - *Day 1 (9 AM):* Begin roof layout and mounting rack installation.
 - *Day 1 (1 PM):* Lunch break.
 - *Day 1 (2 PM):* Mount all panels.
 - *...continue this for all 3 days.*

WEEK 7: MIDTERM BREAK

This week is the midterm break. No new lesson notes.



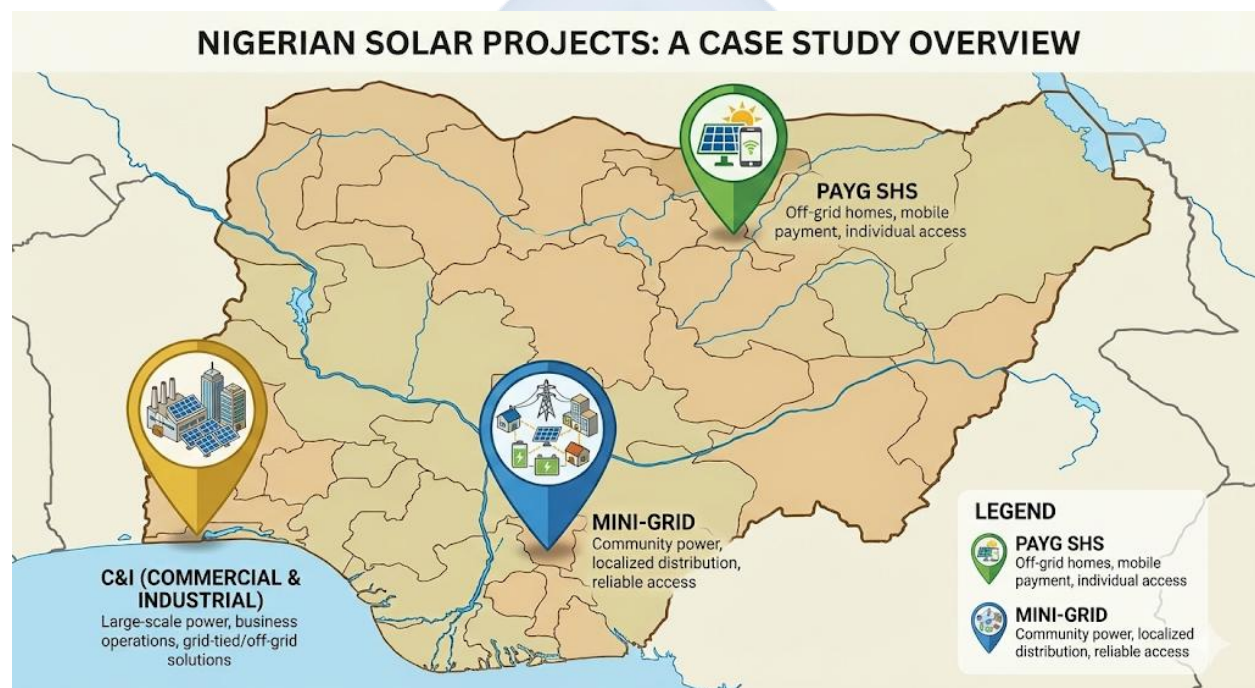
WEEK 8: CASE STUDIES OF NIGERIAN SOLAR PROJECTS

8.1 INTRODUCTION

The solar revolution in Nigeria is already happening at three distinct levels:

1. **PAYG (Pay-As-You-Go):** Small, entry-level systems.
2. **Mini-Grids:** Medium-scale, community-level power.
3. **C&I (Commercial & Industrial):** Large-scale, business-focused power.

Understanding these case studies shows you the different markets you can enter.



8.2 CASE STUDY 1: PAYG SOLAR HOME SYSTEMS (SHS)

- **The Problem:** Over 80 million Nigerians have no access to the grid. They are "off-grid." They spend ~₦150/day on kerosene, candles, and phone charging.
- **The Project:** Companies like **Lumos (MTN)**, **Azuri**, and **d.light**.
- **The Solution:** A small "Solar Home System" kit:
 - A 50W-100W panel.
 - A "Power Box" containing a small battery and charge controller.

- 3-4 DC LED lights.
- A DC USB port for phone charging.
- **The Business Model (PAYG):**
 1. A customer pays a small deposit (e.g., ₦10,000).
 2. They make daily payments (e.g., ₦120/day) via mobile airtime.
 3. If they pay, the system is "unlocked" and works.
 4. If they don't pay, the system locks itself.
 5. After ~3 years, the system is paid off, and they own it.
- **The Impact:**
 - **Social:** Replaces dangerous kerosene (no more fumes/fires). Children can study at night.
 - **Economic:** The daily PAYG cost is *cheaper* than their old "energy bill" (kerosene + charging).

8.3 CASE STUDY 2: RURAL ELECTRIFICATION (MINI-GRIDS)

- **The Problem:** A PAYG system can't power a "productive" business (like a welder, a cold room, or a rice mill). An entire community needs a larger solution.
- **The Project:** The **Rural Electrification Agency (REA)** of Nigeria, via its **Nigeria Electrification Project (NEP)**, funds private companies to build mini-grids.
- **Examples:** Rokota (Niger), Sabon Gari Market (Kano), Sura Shopping Complex (Lagos).
- **The Solution:** A "Mini-Grid" (or "Micro-Grid").
 - A large central solar farm (e.g., 100kWp to 500kWp).
 - A large battery bank (often in a container).
 - A diesel generator for backup.
 - A *local grid* (poles and wires) that connects every house and business in the village.
- **The Business Model:**
 1. A private company (the "developer") builds and operates the grid.
 2. Every user (house, shop) gets a pre-paid meter.

3. Users buy power (e.g., ₦X per kWh) just like they would buy NEPA credit.
 - **The Impact:** True economic transformation. Businesses can run freezers, welders, and mills, creating jobs and increasing income for the entire community.

8.4 CASE STUDY 3: COMMERCIAL & INDUSTRIAL (C&I)

- **The Problem:** A large bank, factory, or university is *on* the grid, but it's not reliable. Their biggest operational cost is *diesel* for their massive generators.
- **The Project:** A "C&I" solar installation (e.g., Access Bank, Nigerian Breweries, many factories).
- **The Solution:** A large **Hybrid** or **Grid-Tied** system.
 - A 100kWp to 1MW+ PV array, usually installed on the factory's large roof ("roof-mount") or as carports.
 - A large inverter/battery system that "blends" solar, grid, and diesel power to find the cheapest, most reliable mix.
- **The Business Model ("Solar-as-a-Service"):**
 - The factory/bank pays **₦0** for the installation.
 - A "solar developer" pays for and builds the entire system.
 - The developer sells the *power* to the factory at a "tariff" (e.g., ₦150/kWh) that is cheaper than their diesel cost (e.g., ₦250/kWh).
 - It's a "win-win": The factory saves money with no risk, and the solar developer gets a long-term (e.g., 20-year) contract.
- **The Impact:** Massive reduction in operational costs, lower carbon footprint, and energy independence for big businesses.

EVALUATION

1. What are the three (3) main levels of solar projects in Nigeria?
2. What is the "PAYG" business model, and who is its target customer?
3. What is a "Mini-Grid"? How is it different from a "Solar Home System"?
4. What is the main *problem* that C&I solar systems solve for banks and factories?
5. What is the "Solar-as-a-Service" business model? Why is it attractive to a factory owner?

ASSIGNMENT

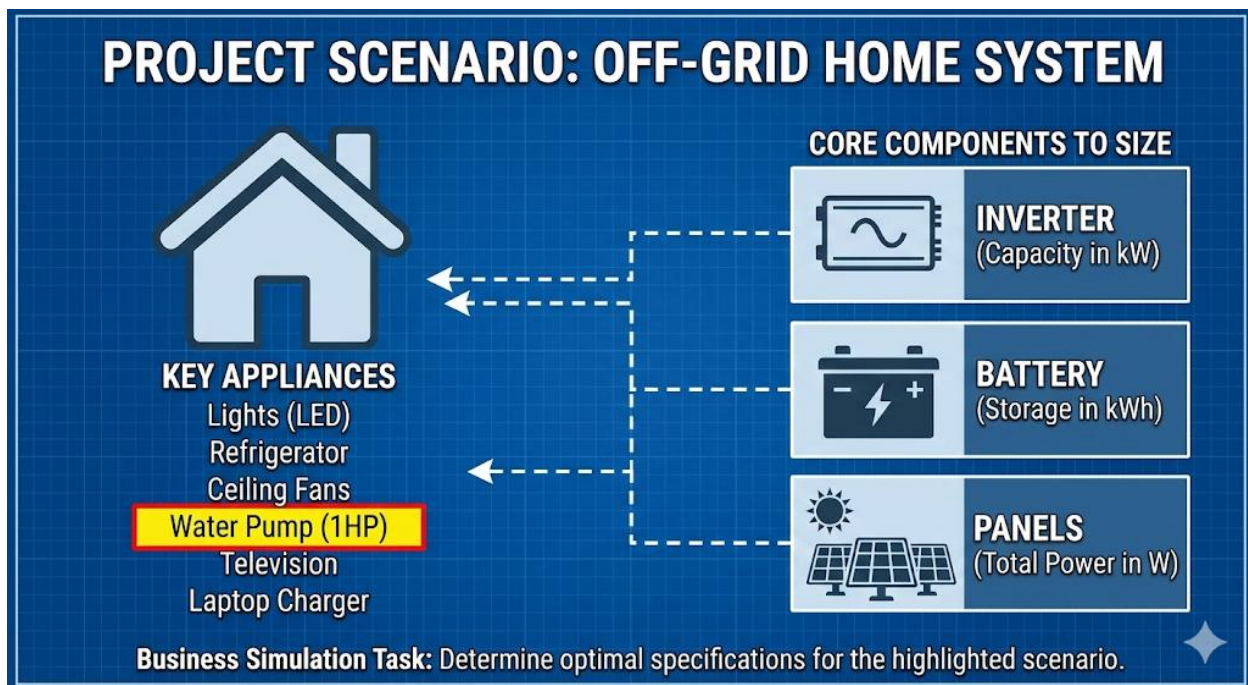
1. The Rural Electrification Agency (REA) is the most important government body for this sector.
2. Go to their website (rea.gov.ng) and find their "Projects" page.
3. Find one *specific* mini-grid project they have completed.
4. Write a half-page summary: What is the name of the project? Where is it located? How many people or businesses does it serve?



WEEK 9: PRACTICAL: BUSINESS SIMULATION (PROPOSAL FOR A 5KVA SYSTEM)

9.1 OBJECTIVE

To synthesize all the technical (Term 1 & 2) and business (Term 3) knowledge into a single, practical project. This is the "capstone" of your SS2 training.



9.2 THE SCENARIO

- **The Client:** Mr. & Mrs. Bolaji, living in a 3-bedroom bungalow in Abeokuta.
- **The Problem:** They have 12-15 hours of grid power, but it's unpredictable. Their main cost is running a 6.5kVA petrol generator for 4-6 hours a day, costing ~₦3,000/day.
- **Their Goal:** They want 24/7 power for their "essential loads" and want to stop using the generator.
- **The Loads (To be powered by Solar):**
 - 1 x 150W Refrigerator
 - 10 x 10W LED Lights
 - 4 x 80W Ceiling Fans

- 2 x 100W Televisions
- 1 x 1HP (745W) Water Pump (used 1 hour/day)
- **The Task:** As a team, prepare a professional, itemized proposal for this client.

9.3 PRACTICAL STEPS (IN GROUPS)

STEP 1: ANALYZE THE LOAD & MANAGE EXPECTATIONS (30 mins)

- First, calculate the *Daily Energy Need (Wh)*.
 - Fridge: $150\text{W} \times 12\text{h (cycle)} = 1,800\text{ Wh}$
 - Lights: $10 \times 10\text{W} \times 6\text{h} = 600\text{ Wh}$
 - Fans: $4 \times 80\text{W} \times 8\text{h} = 2,560\text{ Wh}$
 - TVs: $2 \times 100\text{W} \times 6\text{h} = 1,200\text{ Wh}$
 - Pump: $1 \times 745\text{W} \times 1\text{h} = 745\text{ Wh}$
 - **Total Energy = ~7,000 Wh (7.0 kWh) / day**
- Second, identify the "Problem Load."
 - It's the **Water Pump**. It has a "surge" (inrush) current of 3-5x its running power. A 5kVA inverter *can* start it, but it's a heavy load.
 - **Management Strategy:** You must advise the client: "The system will power your pump, but you *must* use it only during the day (10 AM - 4 PM) when the sun is shining, not at night."

STEP 2: SIZE THE SYSTEM & CHOOSE COMPONENTS (30 mins)

1. **Inverter:** 5kVA / 48V Hybrid Inverter. (Good choice, handles the 745W pump surge).
2. **Battery Bank:** They need 7,000 Wh.
 - *Option 1 (Lithium):* We need to cover 7.0 kWh. A single **10.2 kWh / 48V LiFePO₄** battery (e.g., Pylontech, BYD) is perfect. (80% DoD = 8,160 Wh available).
 - *Option 2 (Lead-Acid):* $(7,000\text{ Wh} \times 1\text{ day}) / (0.50\text{ DoD} \times 48\text{V}) = 291\text{ Ah}$.
 - They would need four (4) 12V 300Ah batteries. This will be *cheaper* but *heavier* and *won't last* as long.

- **Decision:** Propose *both* options and *recommend* the Lithium. We will price the Lithium.
- 3. **PV Array:** Must recharge the 7,000 Wh used, *plus* power daytime loads.
 - $(7,000 \text{ Wh} / 0.80 \text{ efficiency}) / 5 \text{ PSH} = 1,750 \text{ Wp}$.
 - This is the *minimum*. A better sizing is $(\text{Battery kWh} / 2) = 10.2 / 2 = 5.1 \text{ kWp}$.
 - **Decision:** Let's propose **8 x 550W panels = 4,400 Wp (4.4kWp)**. This is a good balance.

STEP 3: CREATE THE BOQ & PRICING (30 mins)

- Use your phones to research *current Nigerian prices* for these items.
- (Use the BOQ from Week 3 as a template).
- *Majors:* 1 x 5kVA Inverter, 1 x 10.2kWh Battery, 8 x 550W Panels.
- *BOS:* Racking, 6mm cable, Combiner Box, Breakers, Earthing.
- *Labor & Logistics.*
- **Calculate your Direct Cost.**
- **Add your Overhead** (e.g., ₦50,000).
- **Add your Profit Margin** (e.g., 20%).
- **Calculate the Final Price (excl. VAT).**

STEP 4: WRITE THE PROPOSAL (30 mins)

- On paper or a laptop, use the 9-part structure from Week 2.
- *Write* the "Executive Summary."
- *Write* the "Understanding the Need" (list the loads).
- *Write* the "Proposed Solution" (list the components you just sized).
- *Write* the "System Performance" (This is where you *manage expectations* about the water pump!).
- *Insert* your BOQ and Final Price.

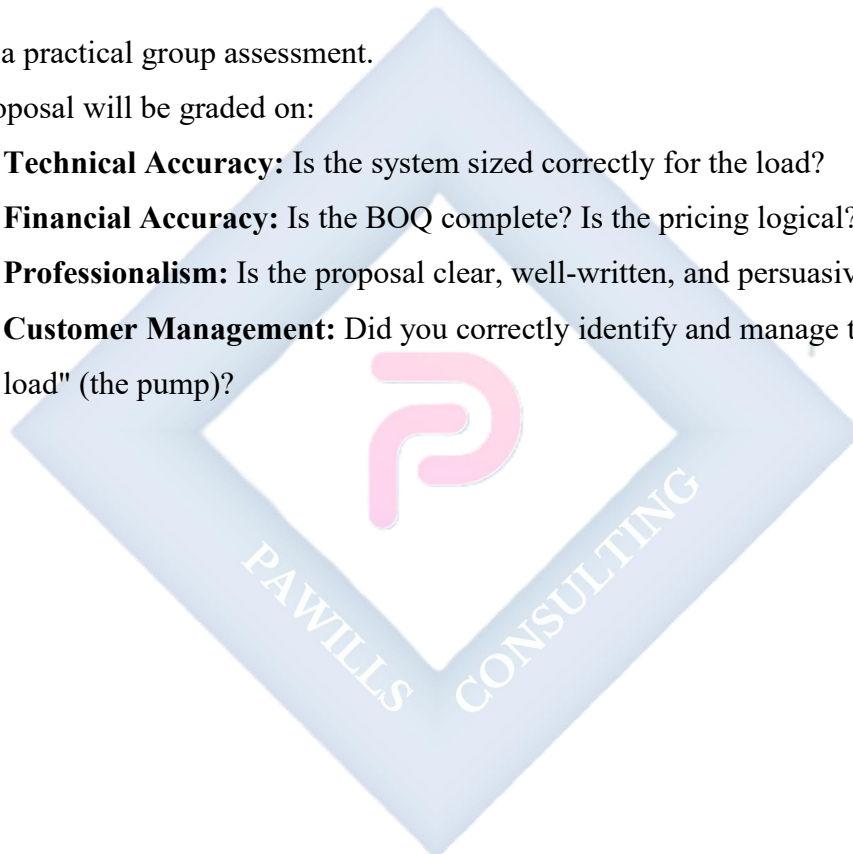
STEP 5: PRESENT THE PROPOSAL (30 mins)

- Each group "presents" their proposal to the instructor (who acts as the client).

- Be ready to answer questions like:
 - "Why is this so expensive? My neighbour did his for ₦1M less." (A: "He likely used lead-acid batteries and fake panels. Our system uses genuine Lithium and will last 10 years.")
 - "Why can't I use my pump at night?" (A: "It's a heavy motor that would drain your entire battery in 2 hours...")

EVALUATION

1. This is a practical group assessment.
2. The proposal will be graded on:
 - **Technical Accuracy:** Is the system sized correctly for the load?
 - **Financial Accuracy:** Is the BOQ complete? Is the pricing logical?
 - **Professionalism:** Is the proposal clear, well-written, and persuasive?
 - **Customer Management:** Did you correctly identify and manage the "problem load" (the pump)?



WEEK 10: FIELD VISIT

10.1 OBJECTIVE

To bridge the gap between classroom theory and the real-world solar industry. This visit is to a professional solar installation company, a solar distributor (wholesaler), or a large-scale solar installation site (e.g., a university, bank, or factory).



10.2 PREPARATION (PRE-VISIT)

1. **Logistics:** All students must have permission, transport, and a set time.
2. **Safety:** Students must wear appropriate attire (e.g., closed-toe shoes) and any PPE required by the host site.
3. **Assignments:** All students must prepare **three (3) questions in advance** to ask the company's staff.
 - *Bad Question:* "Do you sell panels?"
 - *Good Question:* "What is the most common failure point you see in systems installed by amateurs?"

- *Good Question:* "How do you handle procurement to ensure you don't buy sub-standard batteries?"
- *Good Question:* "What advice would you give your younger self when you were first starting in this business?"

10.3 DURING THE VISIT (OBSERVATION)

Students should not just listen; they must *observe* and take notes.

- **Office/Warehouse:** How is their inventory organized? Is it clean and professional? What brands do they stock?
- **Staff:** Who are the people? (Engineers, technicians, sales, admin).
- **Tools:** What professional tools do you see that we have discussed? (e.g., hydraulic crimpers, torque wrenches, multimeters, thermal cameras).
- **On-Site (if applicable):** Observe the team's safety (PPE?), quality of work (trunking, cable management), and professionalism.

10.4 POST-VISIT (THE ASSIGNMENT)

- **Field Visit Report:** Each student must submit a two-page report.
 - **Page 1: Summary:** Where did you go? Who did you speak to? What was the company's business model (e.g., distributor, C&I installer)?
 - **Page 2: Key Learnings:**
 - What were the answers to your three questions?
 - What was the most surprising thing you learned?
 - How has this visit changed your perspective on starting a career in solar?

EVALUATION

1. Submission of the pre-visit questions.
2. Professional conduct during the visit.
3. Submission and quality of the post-visit "Field Visit Report."

WEEKS 11–13: REVISION & EXAMS

11.1 SUMMARY OF TERM 3 (THE BUSINESS OF SOLAR)

This term, we completed the transition from student to professional. We learned:

- **The Opportunity:** Why Nigeria is a prime market for solar (Week 1).
- **How to Get Work:** Writing professional **Proposals** and navigating formal **Tenders** (Week 2).
- **How to Make Money:** The critical difference between **Costing** (BOQ, Overhead) and **Pricing** (Profit Margin) (Week 3).
- **How to Keep Work:** The art of **Customer Service**, managing expectations, and creating recurring revenue through **Maintenance Contracts** (Week 4).
- **How to Manage Work:** Using **Project Management** (Time, Cost, Scope) to deliver jobs professionally (Week 6).
- **The Real World:** Analyzing **Case Studies** (PAYG, Mini-Grids, C&I) and simulating a real **Proposal** (Weeks 8 & 9).

11.2 FINAL EXAM PREPARATION

- The final exam will be weighted heavily on the "Business Simulation." Be prepared to create a simple BOQ and pricing structure for a given load.
- Review all key definitions: Overhead, Profit Margin, BOQ, PAYG, Mini-Grid, C&I, Project Management Triangle.
- Be ready to list the main sections of a professional proposal.
- Be able to explain why customer service (especially managing expectations) is critical to success.